



Chapter 4

扩散

Diffusion of dopants

李秀妍

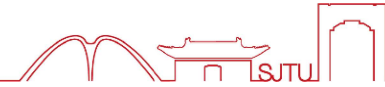
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2025年11月20日



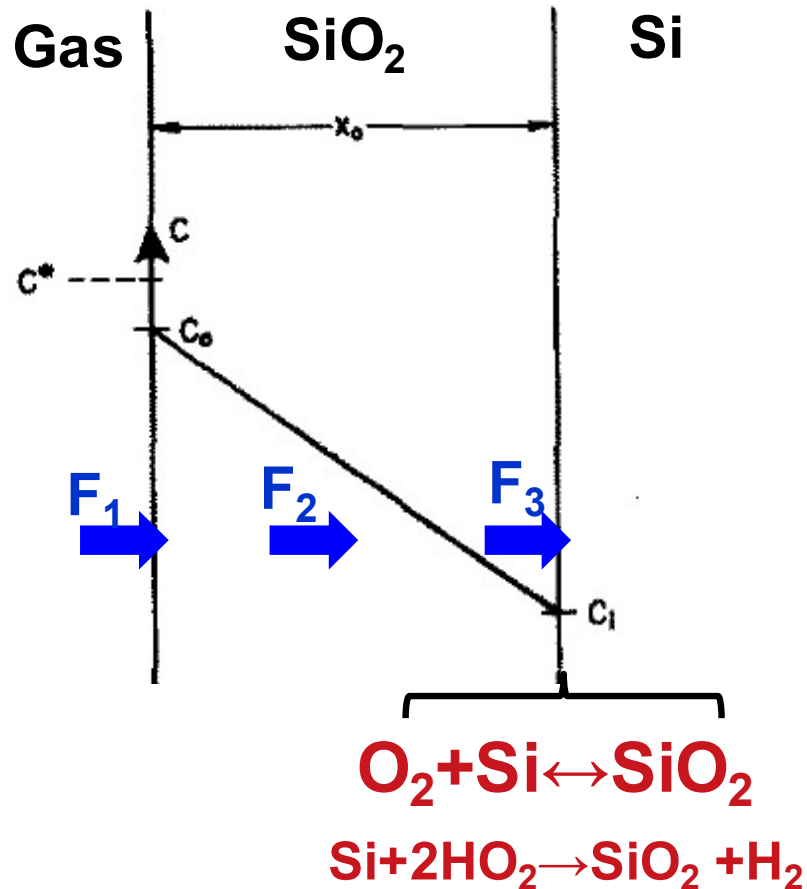
- ✎ **4. 1 Why diffusion is important?**
- ✎ **4. 2 Basics of Diffusion**
- ✎ **4. 3 Fick's law and solution for diffusion**
- ✎ **4. 4 High concentration diffusion effect**
- ✎ **4. 5 Other aspect of diffusion**
- ✎ **4.6 Summary and homework**

4.1 Why is diffusion important



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F: flux – the number of oxygen molecules that crosses a plane per unit area per second



At surface

$$F_1 = h(C^* - C_0) \quad h_g \text{ 是质量输运系数}$$

h: Mass transfer coefficient [cm/sec].

In SiO_2 D_{eff} : Diffusivity [cm²/sec]

$$F_2 = D_{\text{eff}} \frac{(C_0 - C_i)}{x} \quad \text{菲克第一定律}$$

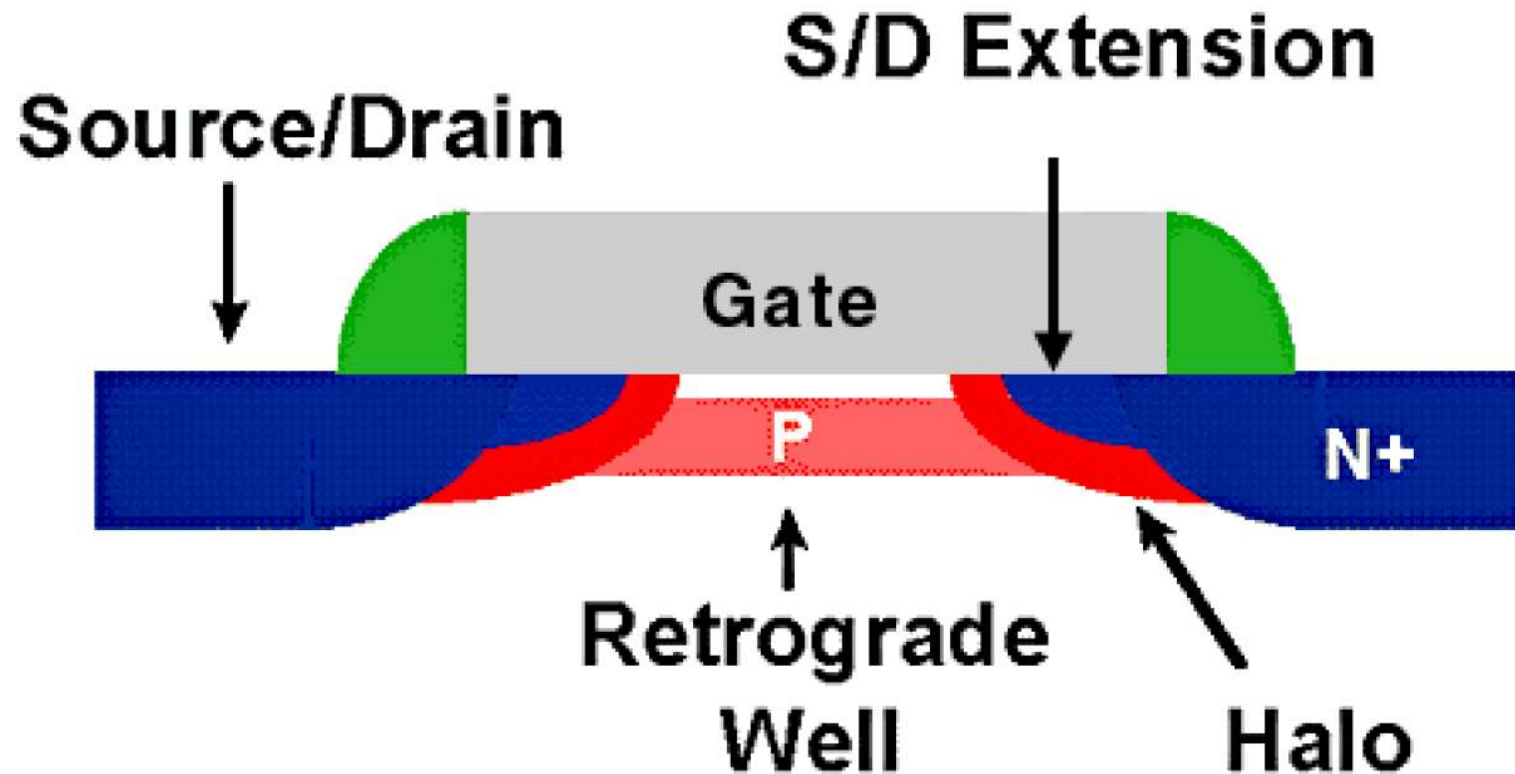
At SiO_2/Si interface

$$F_3 = kC_i \quad \text{k: Oxidation reaction rate constant}$$

Oxidant diffusion → interface reaction.

4.1 Why is diffusion important?

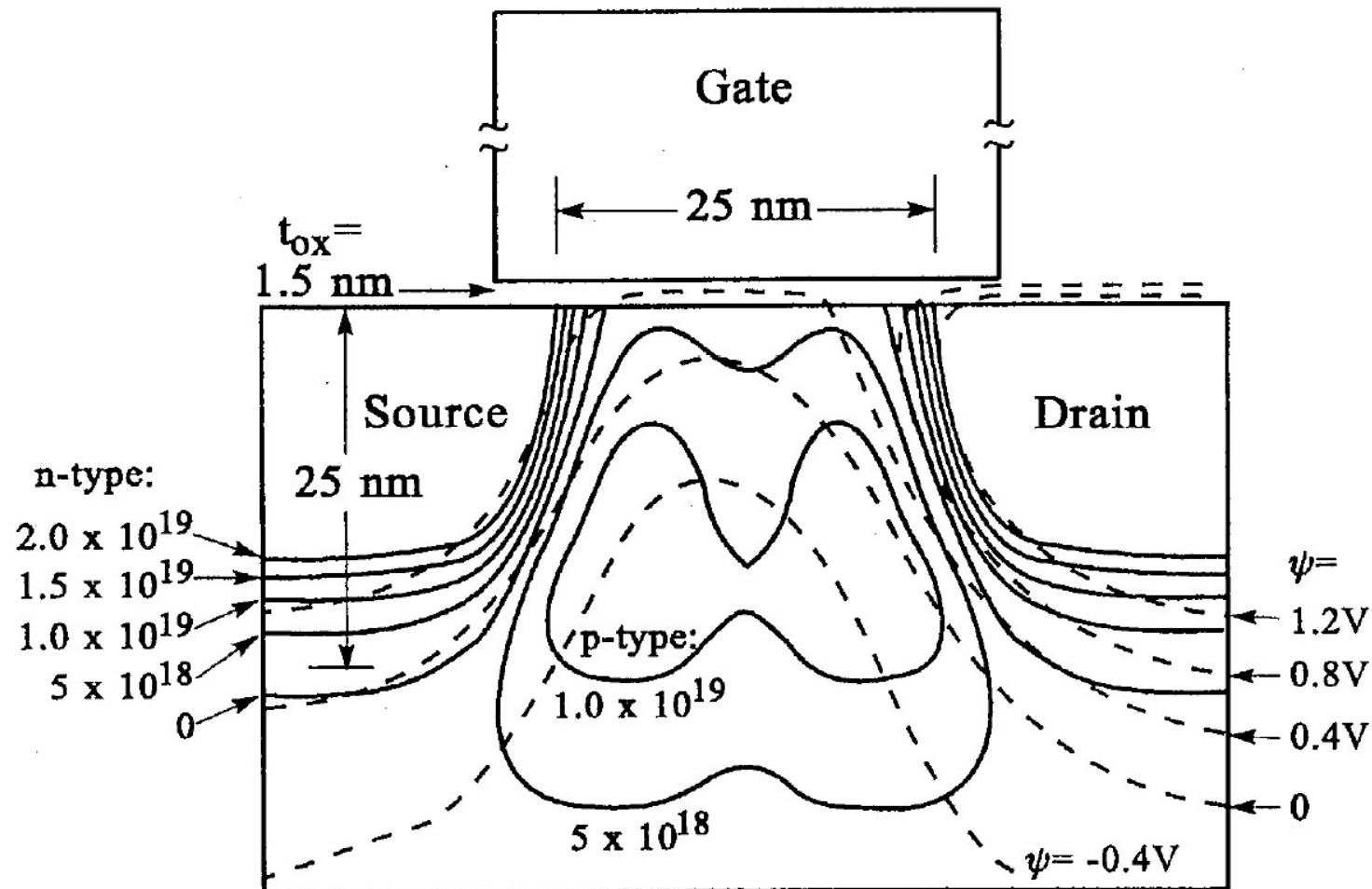
Strategies to overcome short channel effect



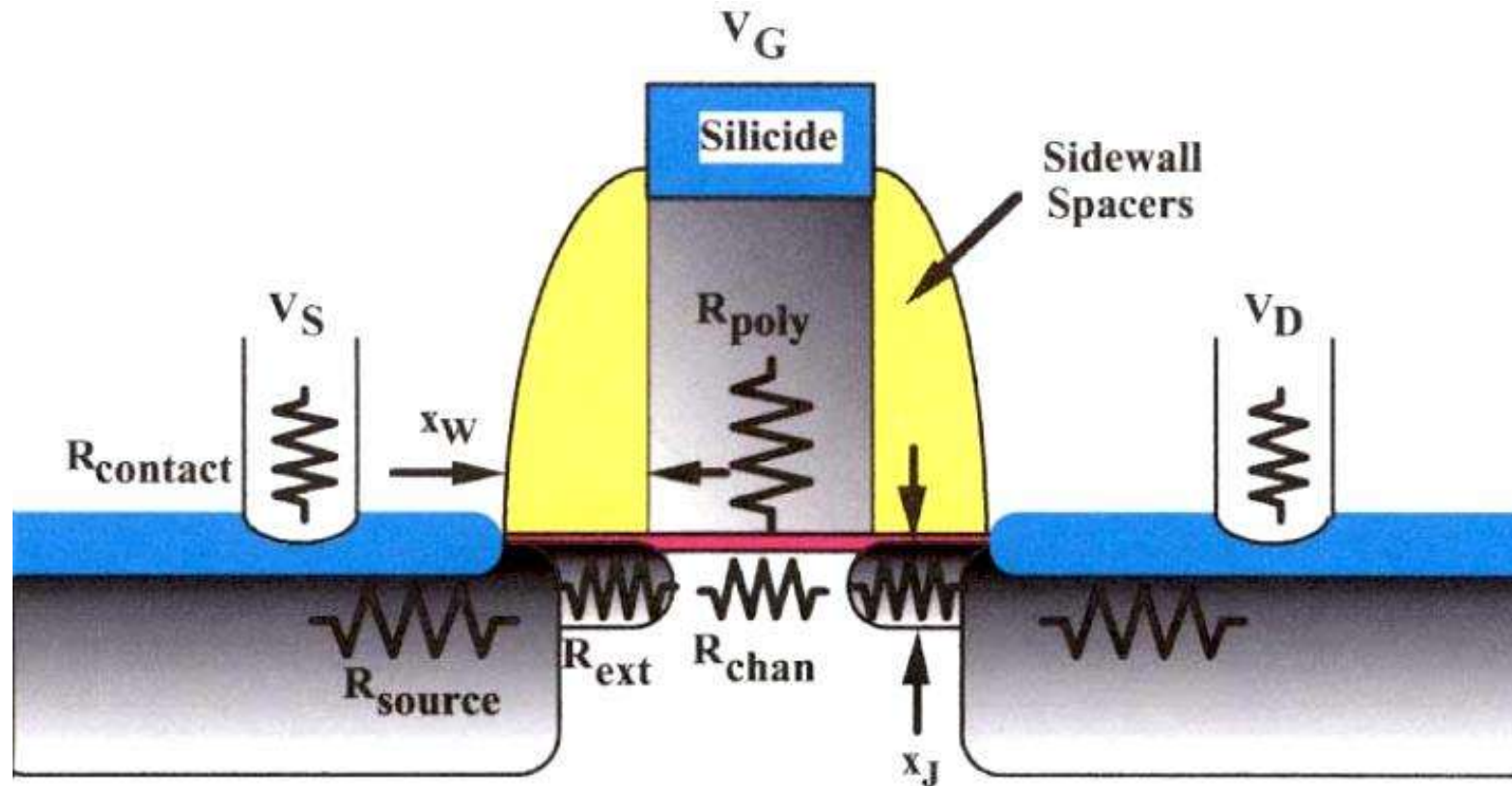
- Anti-punchthrough implantation (PTI)
- Halo implantation

Halo or Pocket Profiles: Improve Manufacturability

25 nm Gate Length MOSFET Design



4.1 Why is diffusion important?



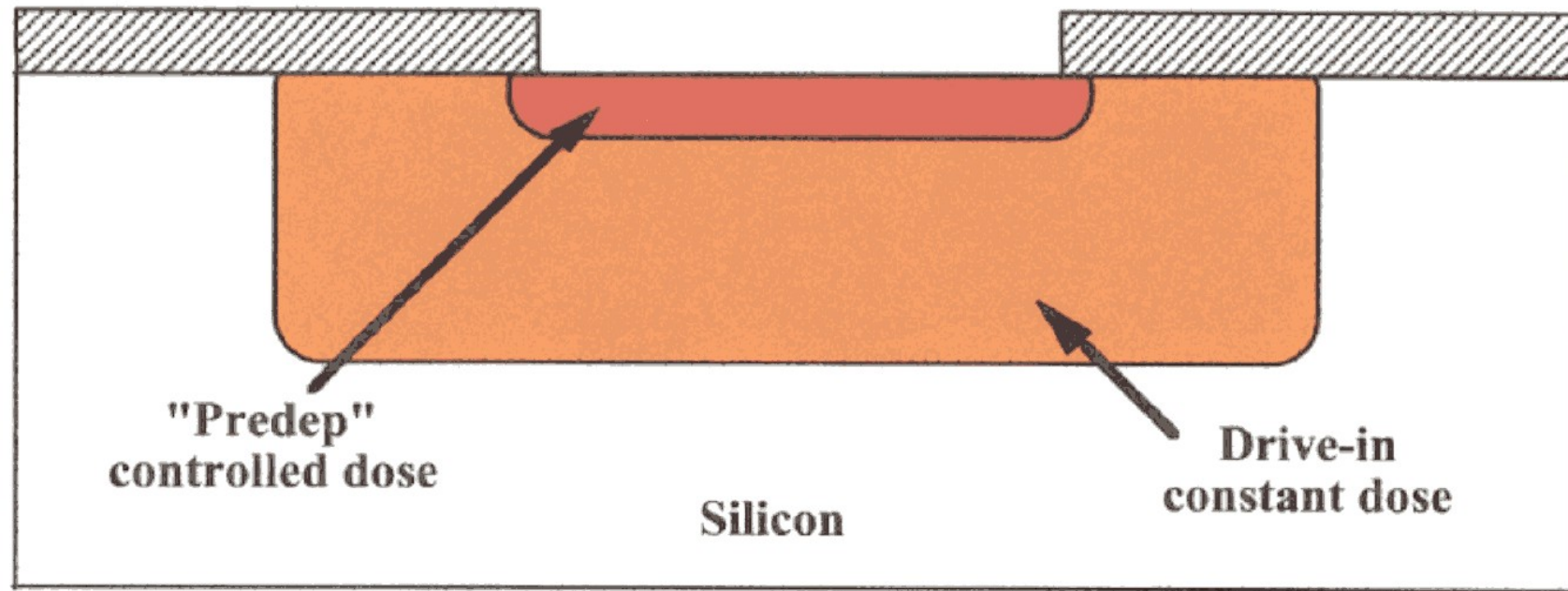
Technology requirements demand knowledge of dopant positions with almost atomic-scale accuracy, in 2D and 3D profiles.

It is based on the knowledge of diffusion

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4. 2.1 How does diffusion occur?

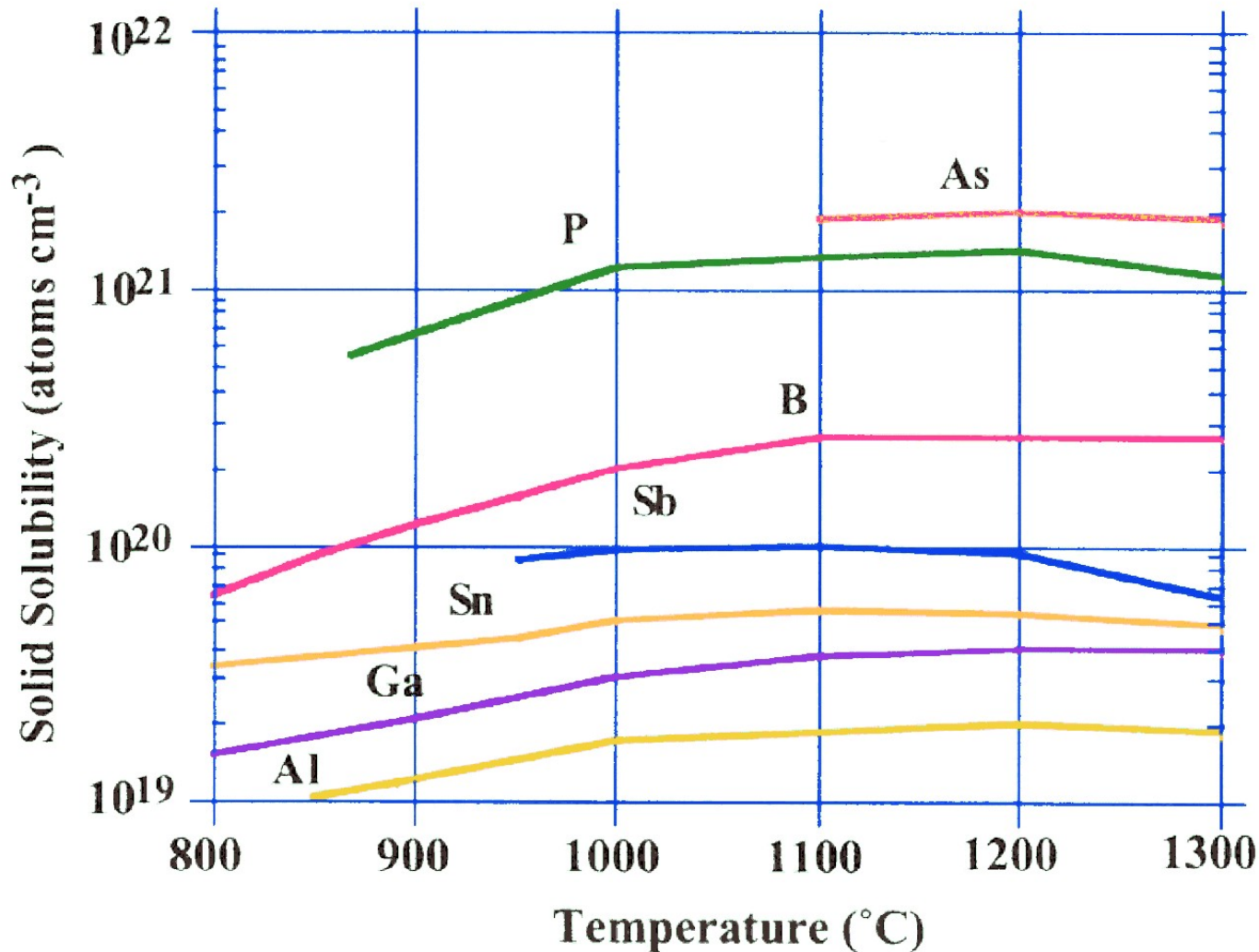
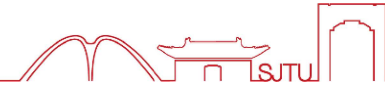
Diffusion is the redistribution of atoms from regions of high concentration of mobile species to regions of low concentrations



Historical development: original method of introducing dopants into the silicon crystal was by doing a gas-phase diffusion (high temperature) masked by oxide. This method has serious limitations on the control of the dose of dopant introduced, as well as the profile shape.

Modern method: use ion implantation to introduce dopants, and annealing/diffusion to repair damage and, if necessary, diffuse the profile.

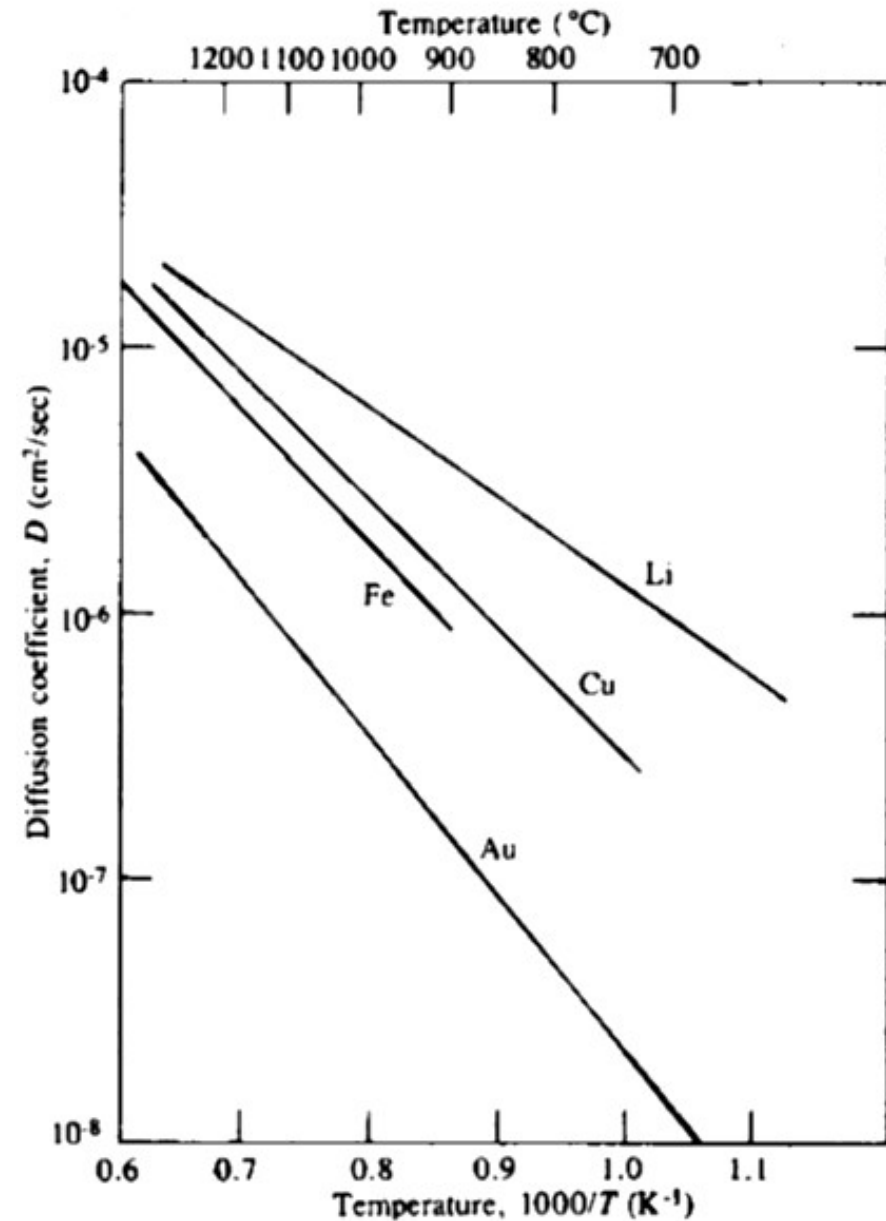
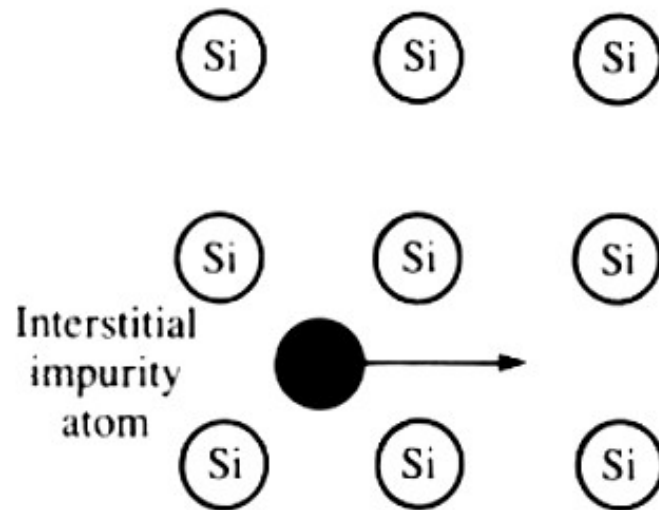
4. 2.1 Solid solubility of dopants



Dopants are soluble in bulk silicon up to a maximum value before they precipitate into another phase (“solid solubility”).

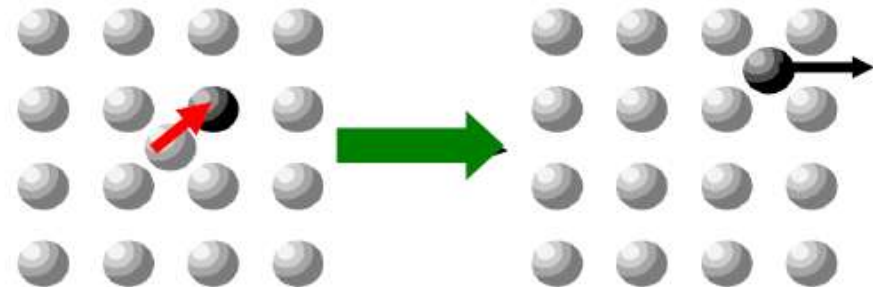
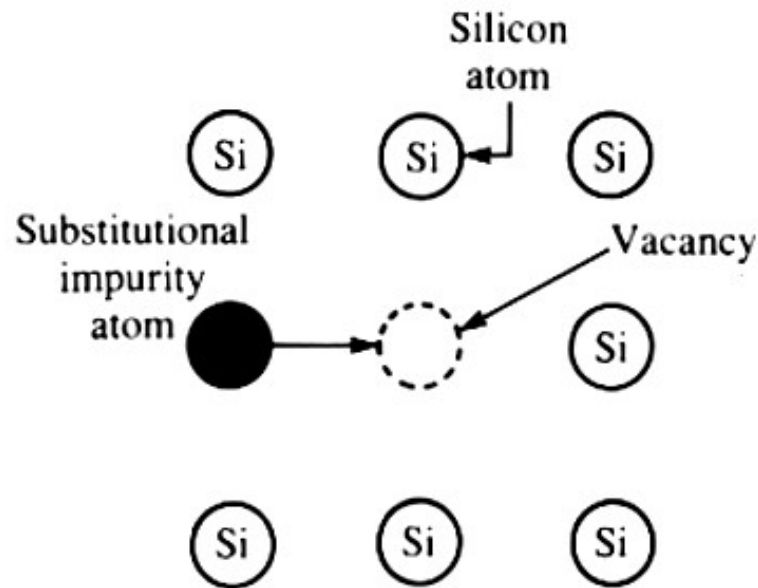
(1) Interstitial Diffusion

Example: Cu, Fe, Li, H



(b) Substitutional Diffusion (c) Interstitialcy Diffusion

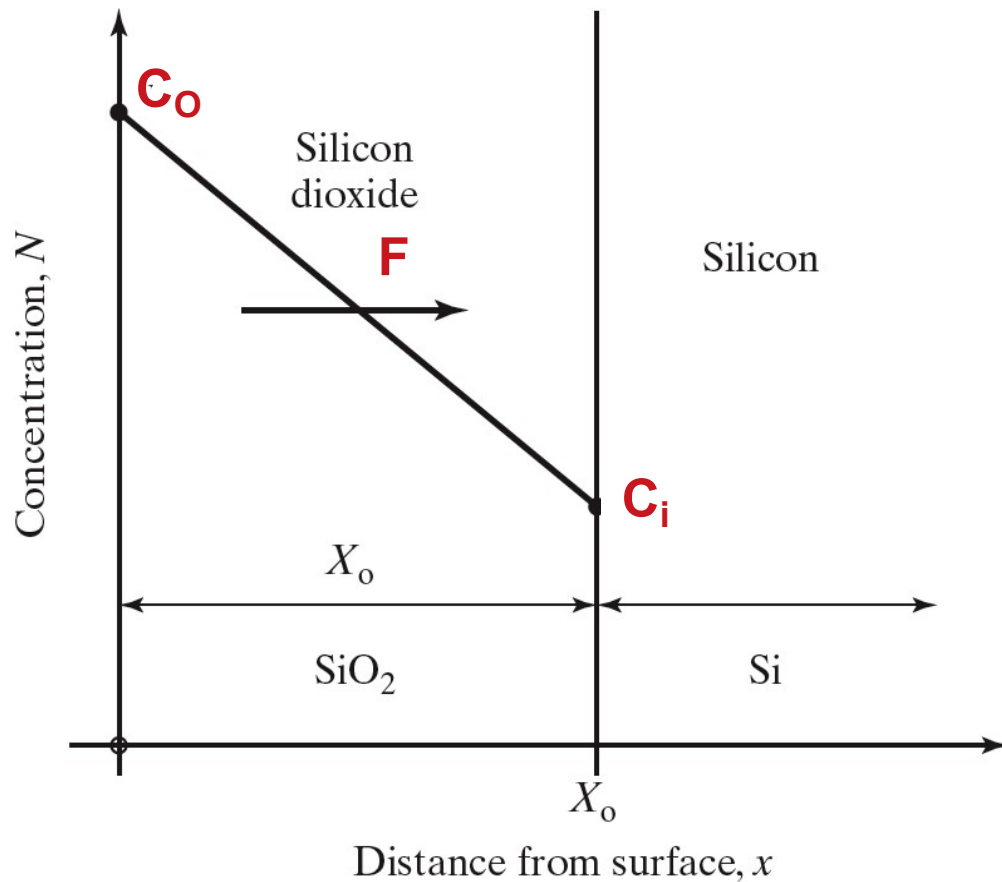
Example: Dopants in Si (e.g. B, P,As,Sb)



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Fick's first law:

How the flux of dopant depends upon the doping gradient:



$$F = - \frac{DdC}{dx}$$

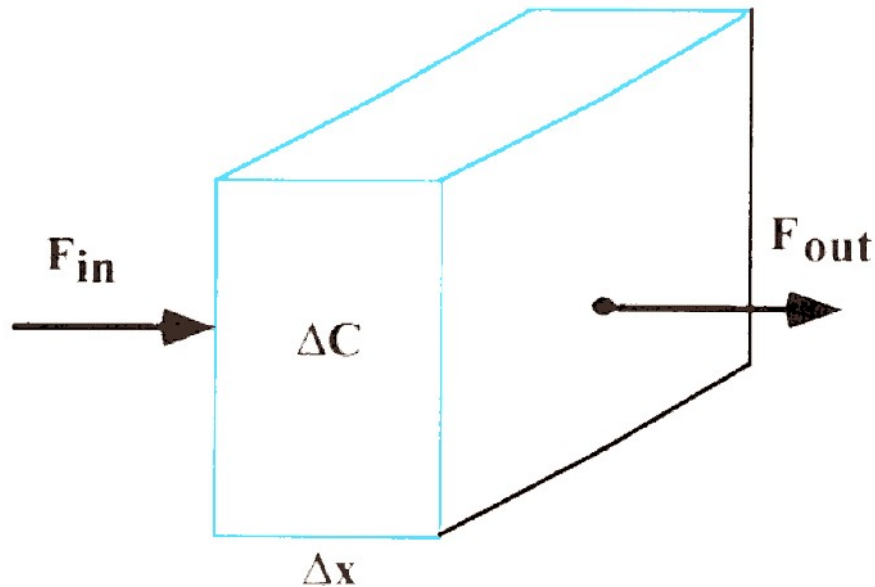
D = diffusion coefficient

Flux– the number of atoms that crosses a plane per unit area per second

When concentration gradient goes to zero, the flow stops.

Fick's Second Law

how the change in concentration in a volume element is determined by the change in fluxes in and out of the volume.

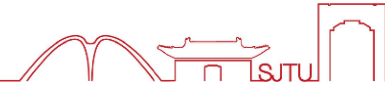


$$\frac{\Delta C}{\Delta t} \Delta x = -(F_{out} - F_{in})$$

$$\frac{\partial C}{\partial t} = -\frac{\partial F}{\partial x} = \frac{\partial}{\partial x} \left(D \frac{\partial C}{\partial x} \right)$$

$$\frac{dC}{dt} = \frac{D d^2 C}{dx^2}$$

4.3.2 Dopant diffusion coefficients



- Related to the atomic hop rate over an energy barrier and is exponentially activated.

$$D = D^0 \exp\left(\frac{-E_A}{kT}\right)$$

Arrhenius Relationship

E_A : activation energy

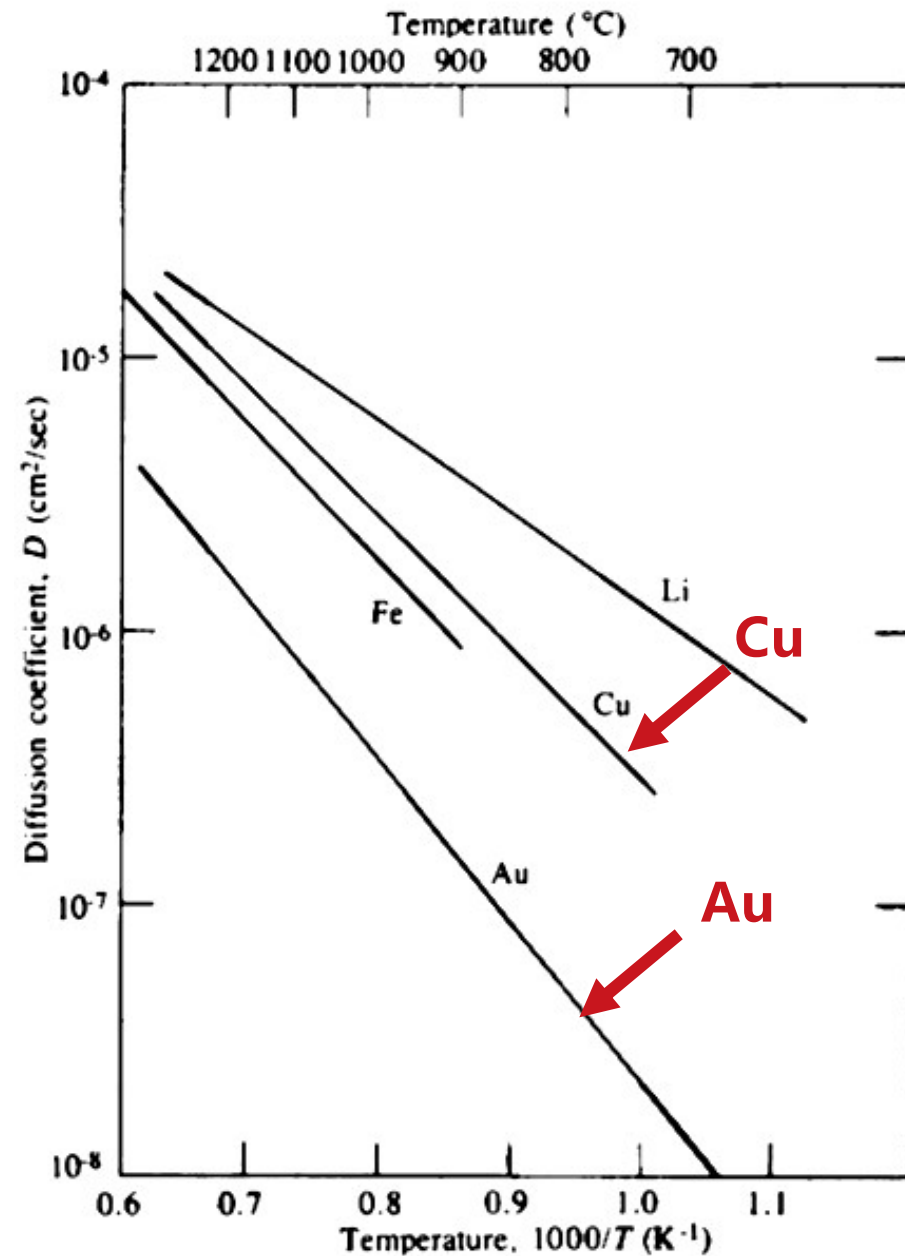
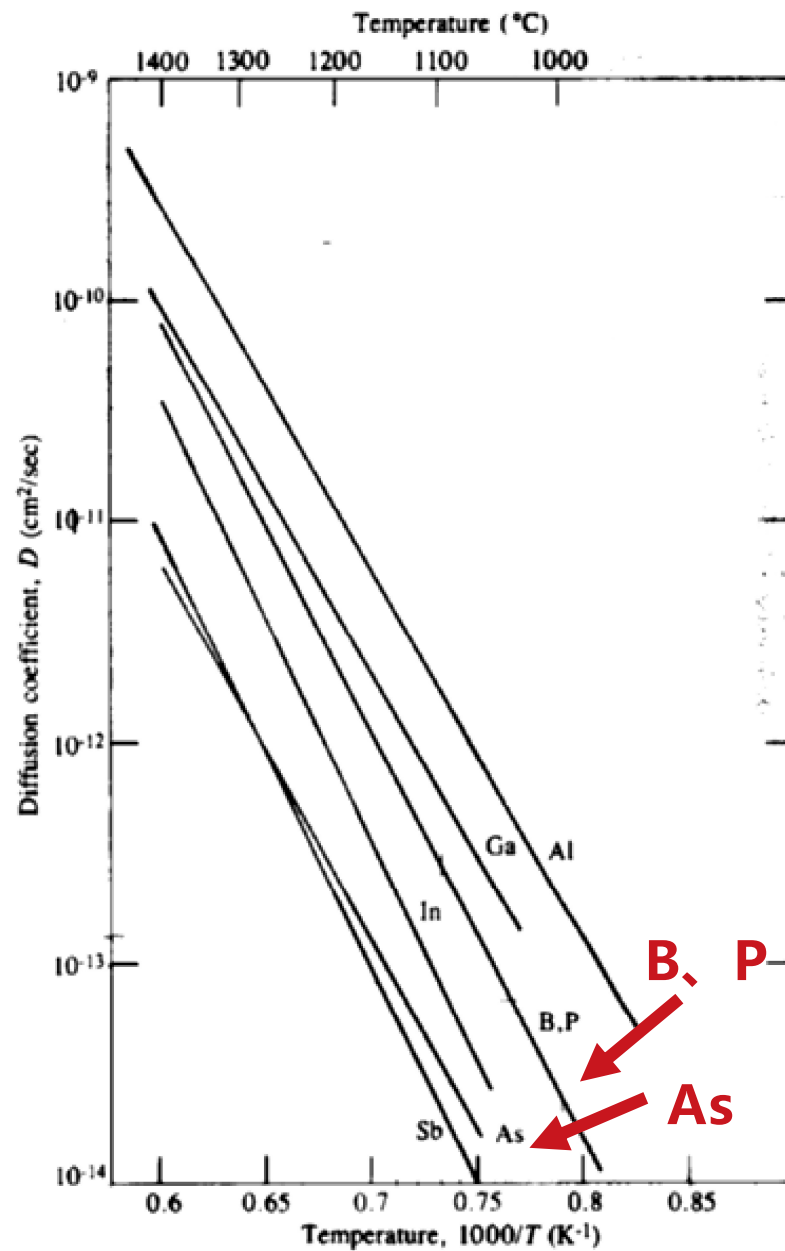
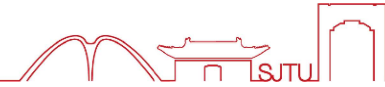
k : Boltzmann's constant

T : absolute temperature

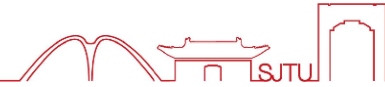
TABLE 4.1 Typical Diffusion Coefficient Values for a Number of Impurities.

Element	$D_0(\text{cm}^2/\text{sec})$	$E_A(\text{eV})$
B	10.5	3.69
Al	8.00	3.47
Ga	3.60	3.51
In	16.5	3.90
P	10.5	3.69
As	0.32	3.56
Sb	5.60	3.95

4.3.2 Dopant diffusion coefficients



4.3.3 Two step Dopant Diffusion

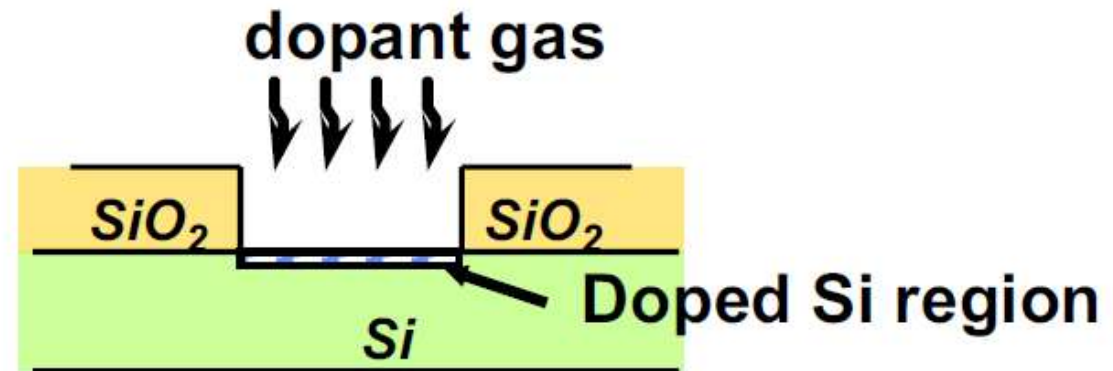


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IC fabrication processes involve a number of different steps

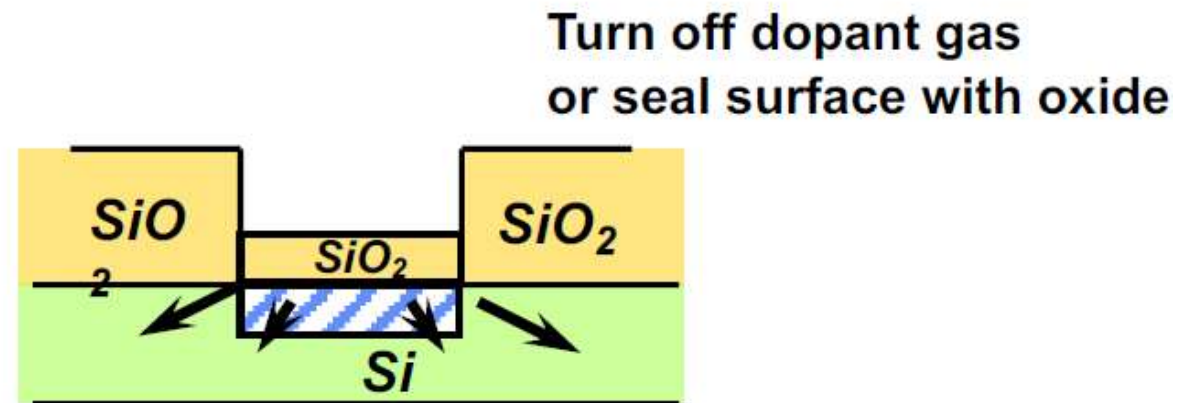
(1) Predeposition

dose control



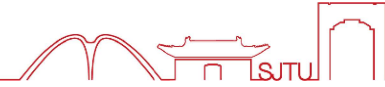
(2) Drive-in

profile control
(junction depth;
concentration)

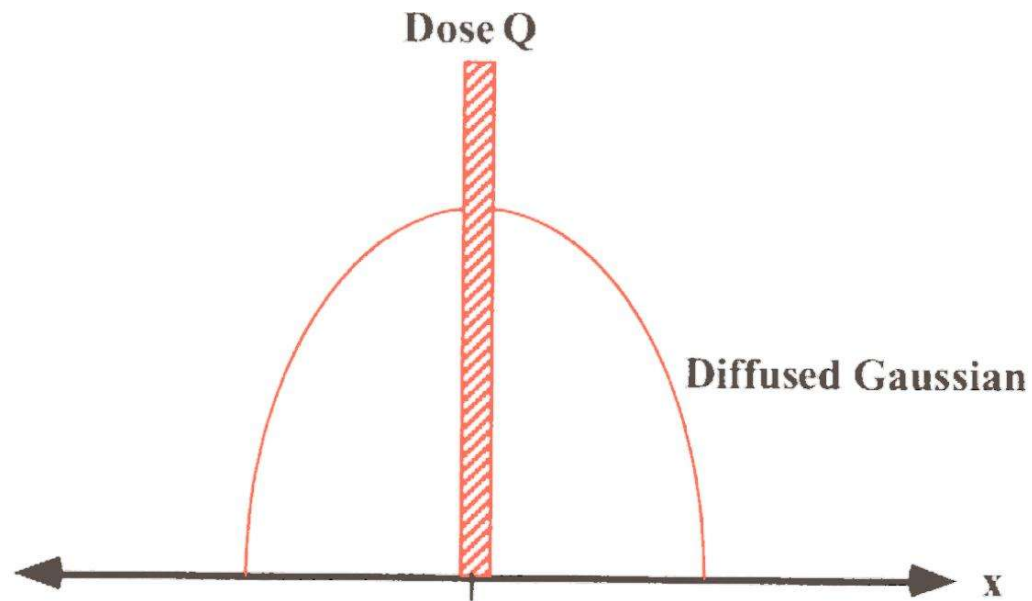


Note: Predeposition by diffusion can be replaced by a shallow implantation step.

4.3.4(1) Constant does



Consider a fixed dose Q , introduced as a delta function at the origin



Boundary condition:

$C \rightarrow 0$ at $t \rightarrow 0$ for $x > 0$

$C \rightarrow \infty$ at $t \rightarrow 0$ for $x = 0$

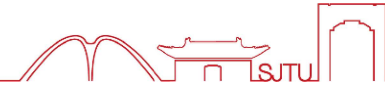
$$\int C(x, t) dx = Q$$

Delta function

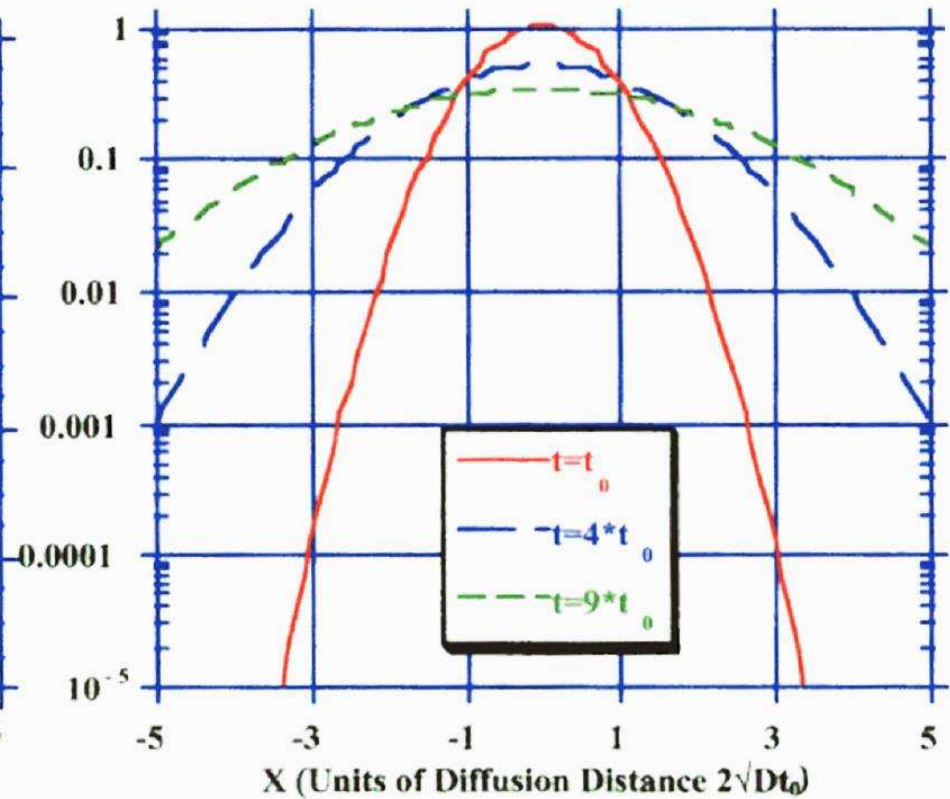
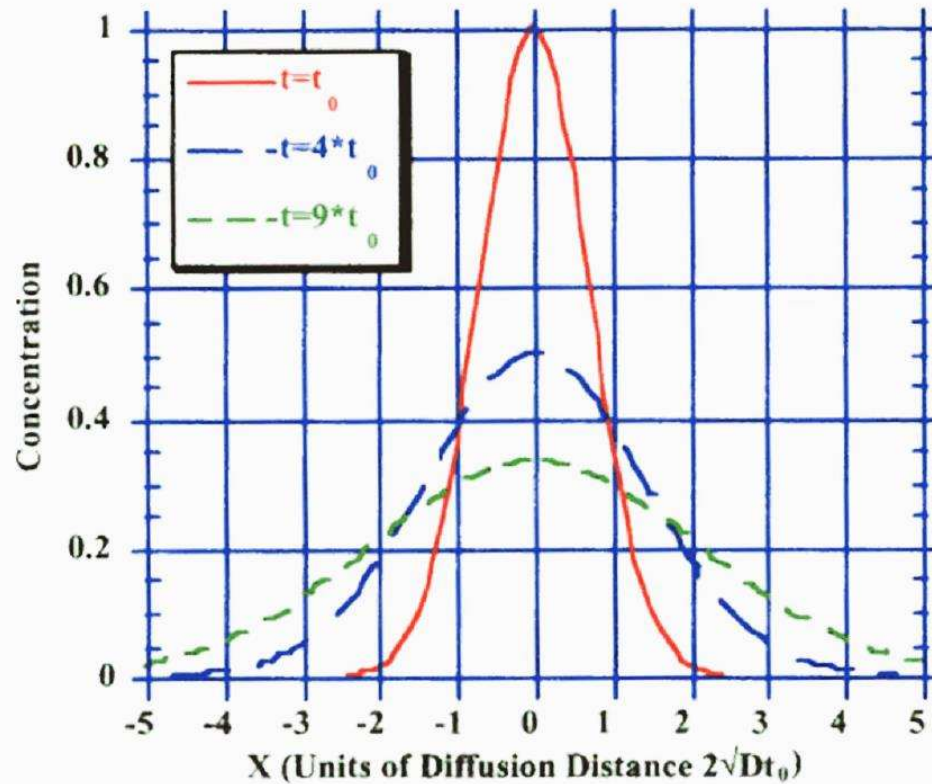
$$C(x, t) = C_0 \exp \left[- \left(\frac{x}{2\sqrt{Dt}} \right)^2 \right] = \frac{Q}{2\sqrt{\pi Dt}} \exp \left(- \frac{x^2}{4Dt} \right)$$

C_0 : Surface Concentration

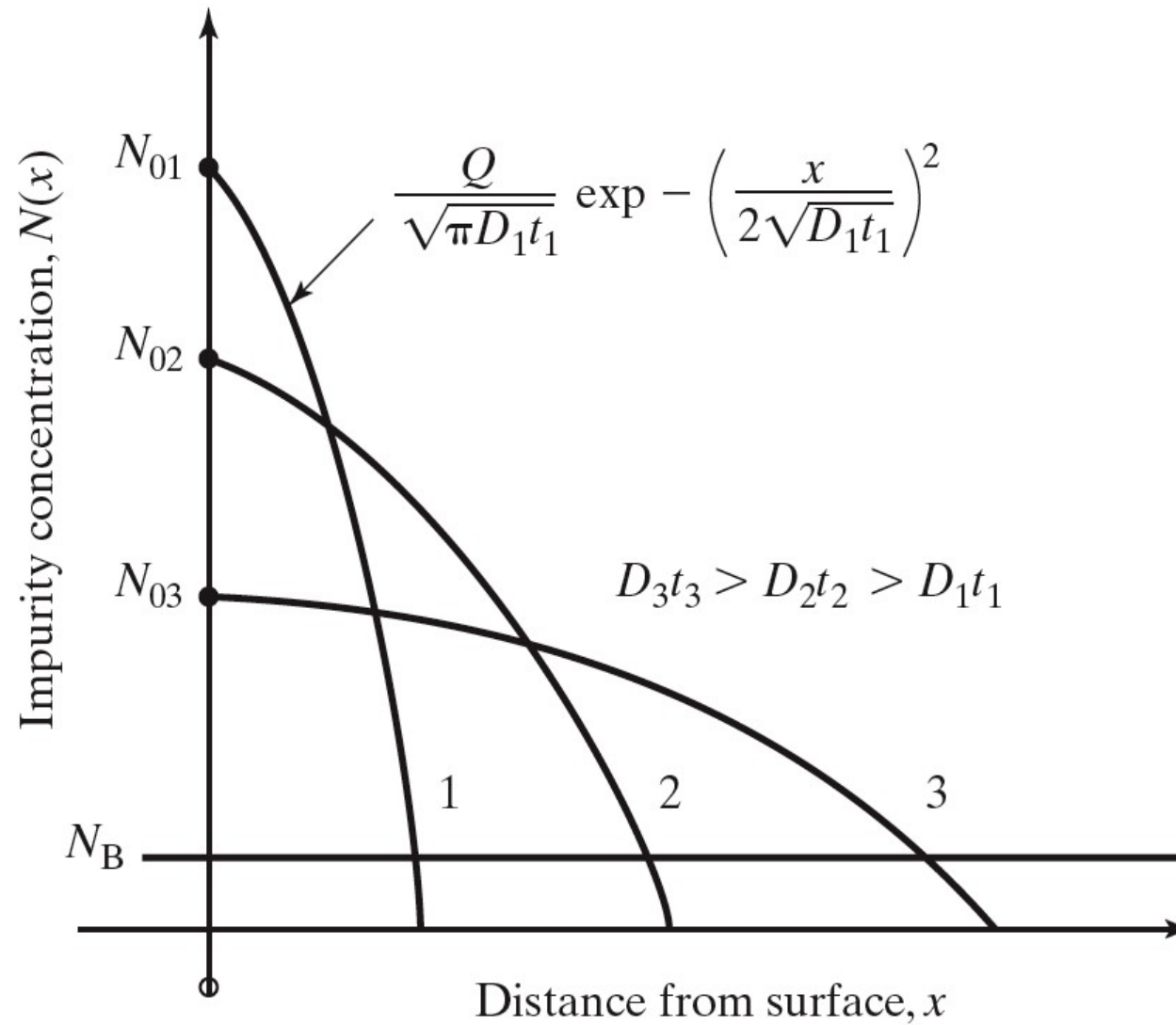
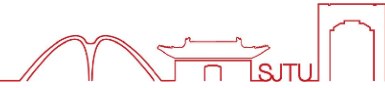
4.3.4(1) Gaussian Diffusion



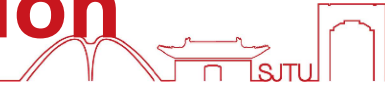
Time evolution of a Gaussian diffusion profile



4.3.4(1) Gaussian Diffusion

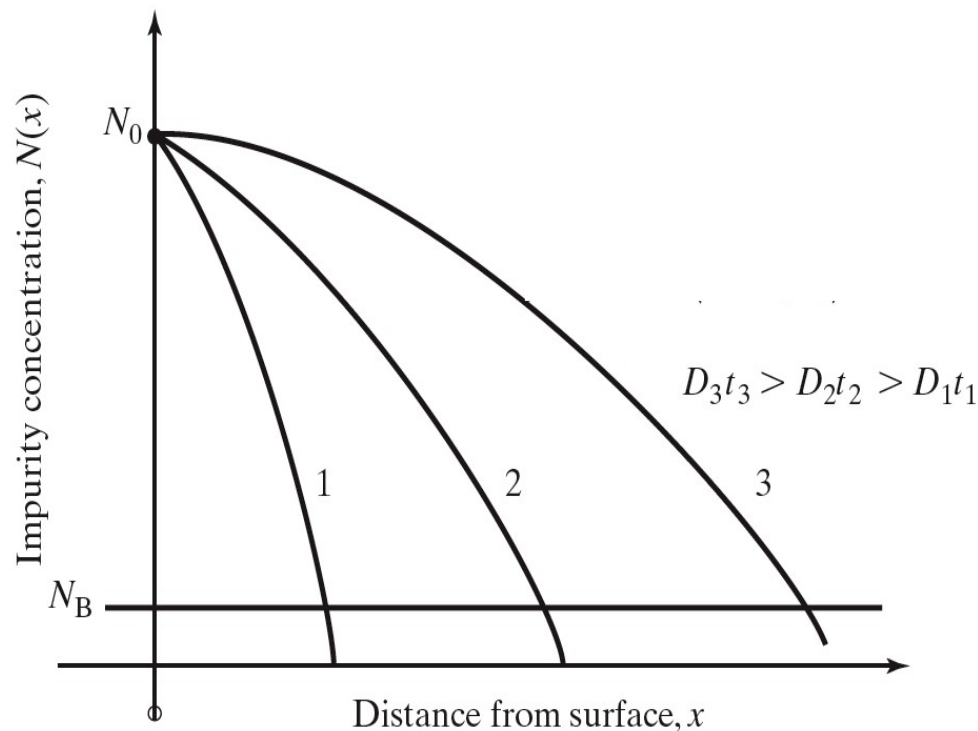


4.3.4(2) Constant surface concentration



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Example: diffusion from a gas ambient into the solid, with the gas concentration above the solid solubility of the dopant



$$C(x, t) = C_0 \operatorname{erfc} \left(\frac{x}{2\sqrt{Dt}} \right)$$

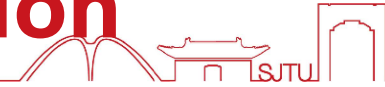
C_0 : Surface concentration

D : Diffusion coefficient

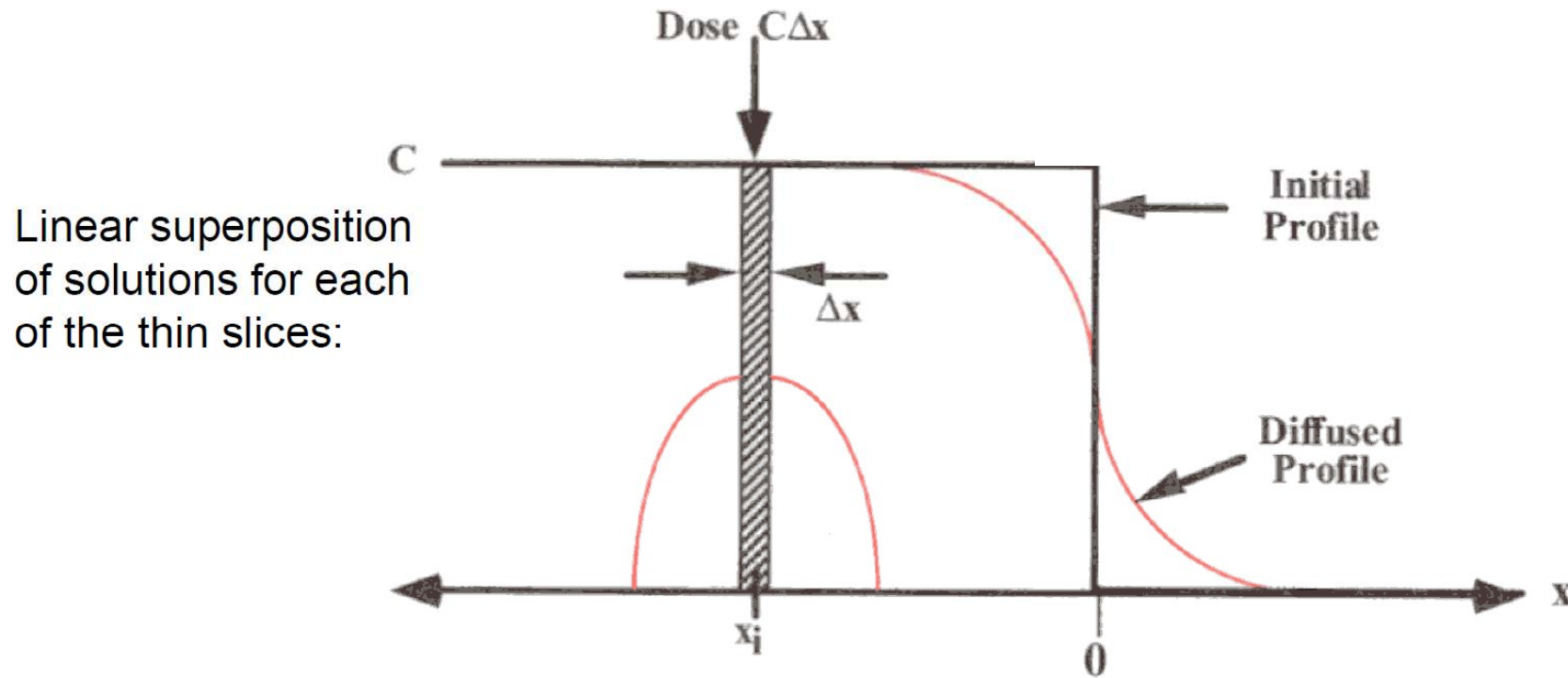
**erfc: Complementary
Erro function**

$$Q = \int_0^{\infty} C(x, t) dx = 2 C_0 \sqrt{\frac{Dt}{\pi}}$$

4.3.4(2) Constant surface concentration



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$$C(x, t) = \frac{Q}{2\sqrt{\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right)$$

$$C(x, t) = \frac{C}{2\sqrt{\pi Dt}} \sum_{i=1}^n \Delta x_i \exp\left(-\frac{(x - x_i)^2}{4Dt}\right)$$

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp(-\eta^2) d\eta$$

Erro function solution is made up of sum of Gaussian delta function solution

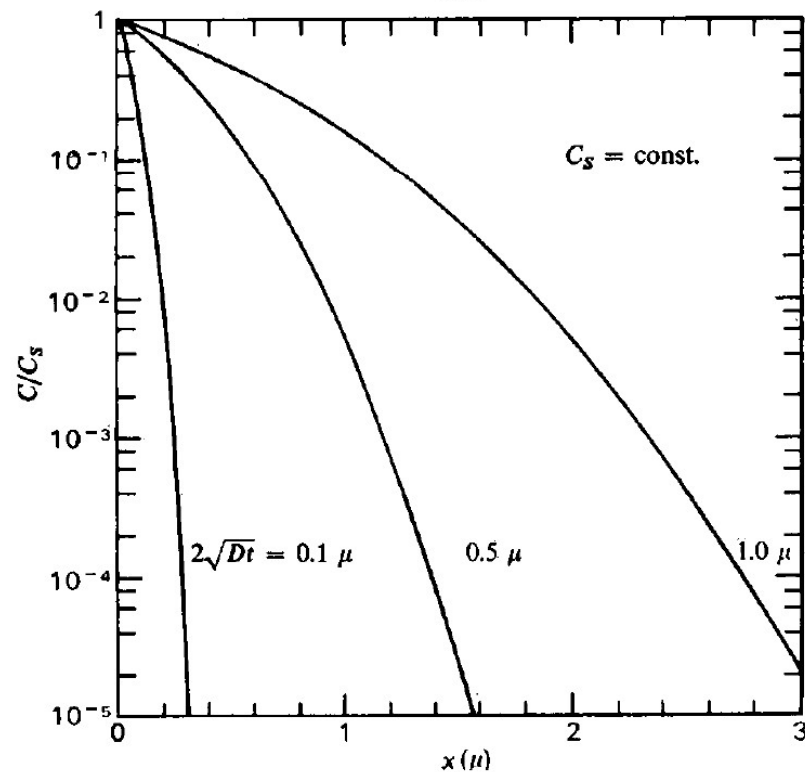
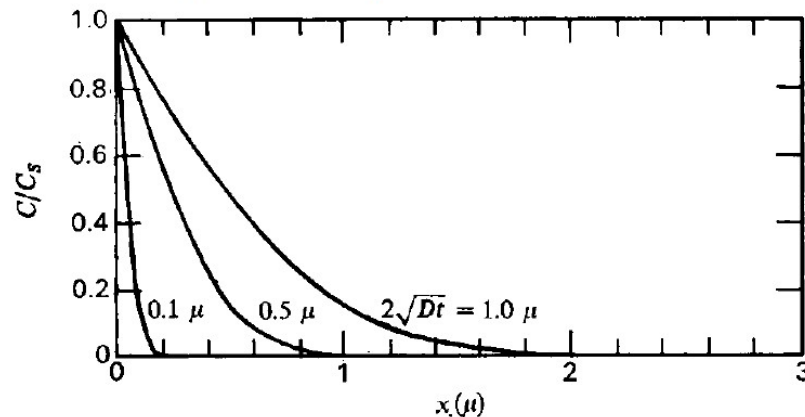
$$C(x, t) = C_0 \text{erfc}\left(\frac{x}{2\sqrt{Dt}}\right)$$

4.3.4 (3) Two kinds of diffusion comparison



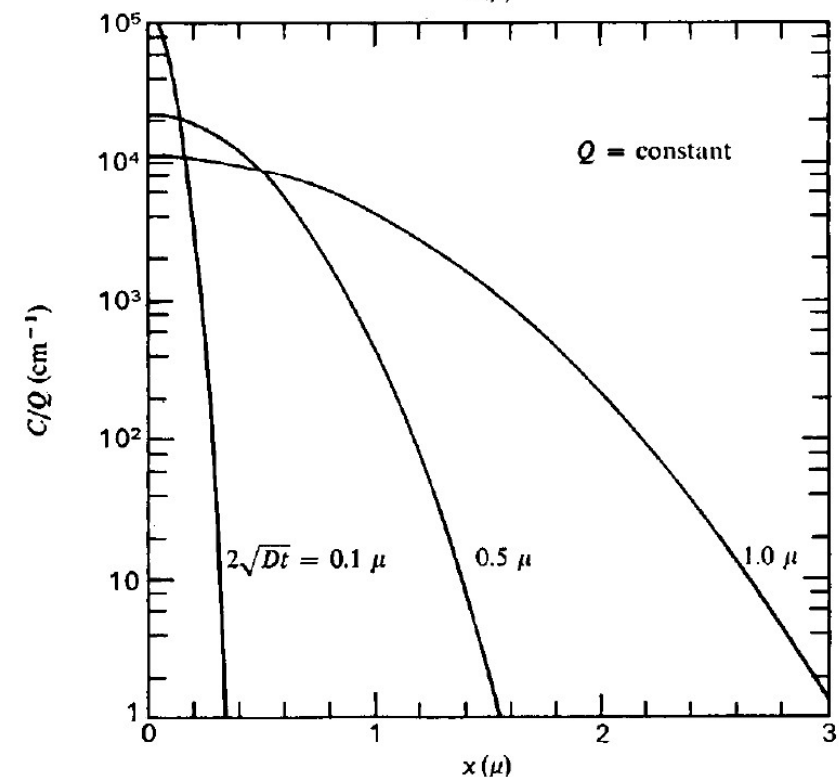
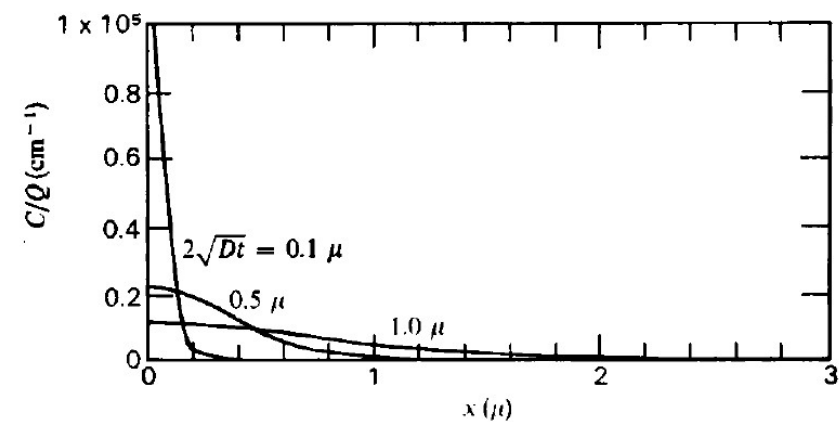
23

Constant surface conc.



erfc

Constant dose Q

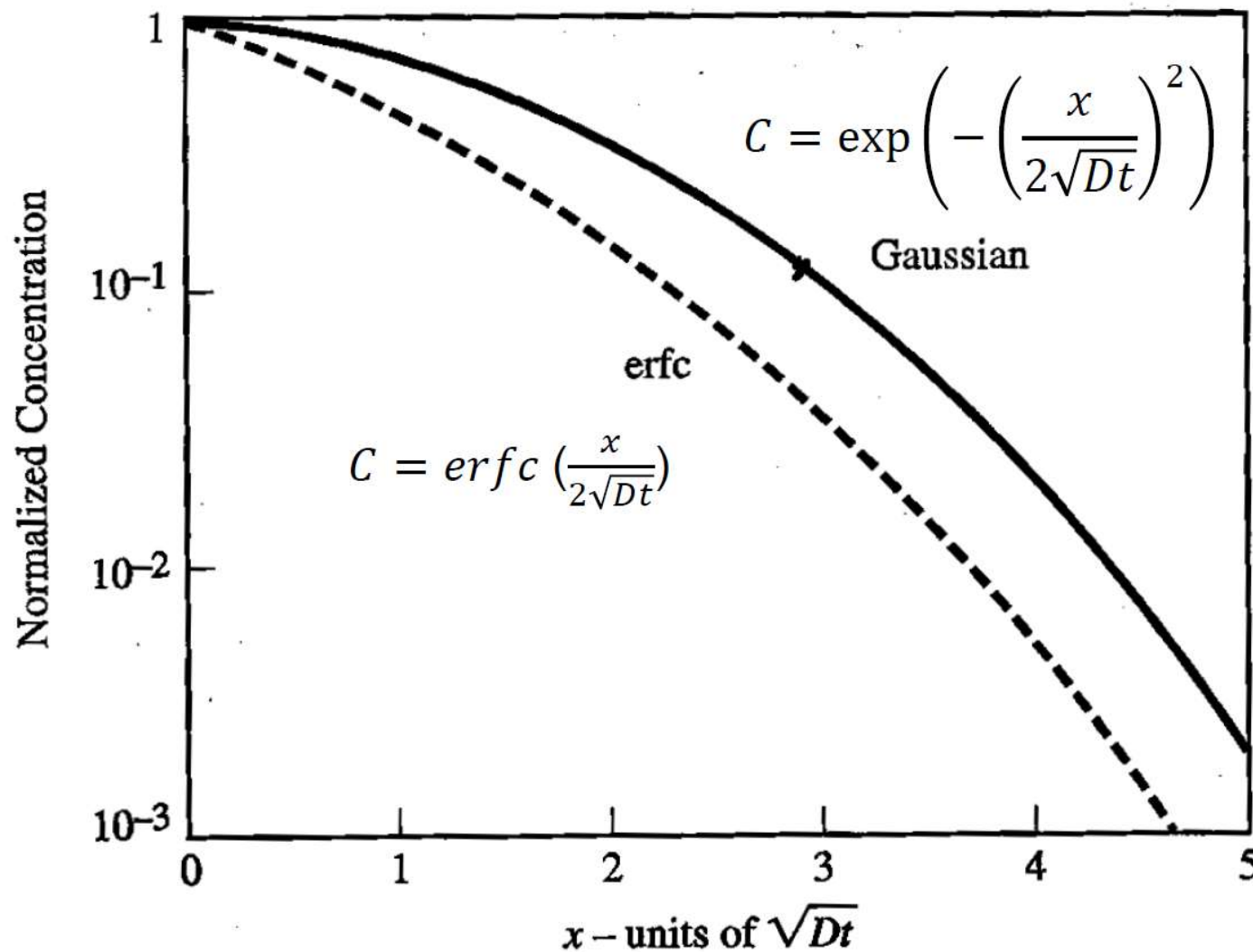


Gaussian

4.3.4 (3) Two kinds of diffusion comparison

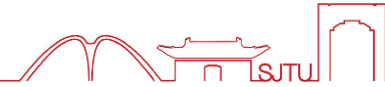


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- Similar shape when normalized to the same surface concentration

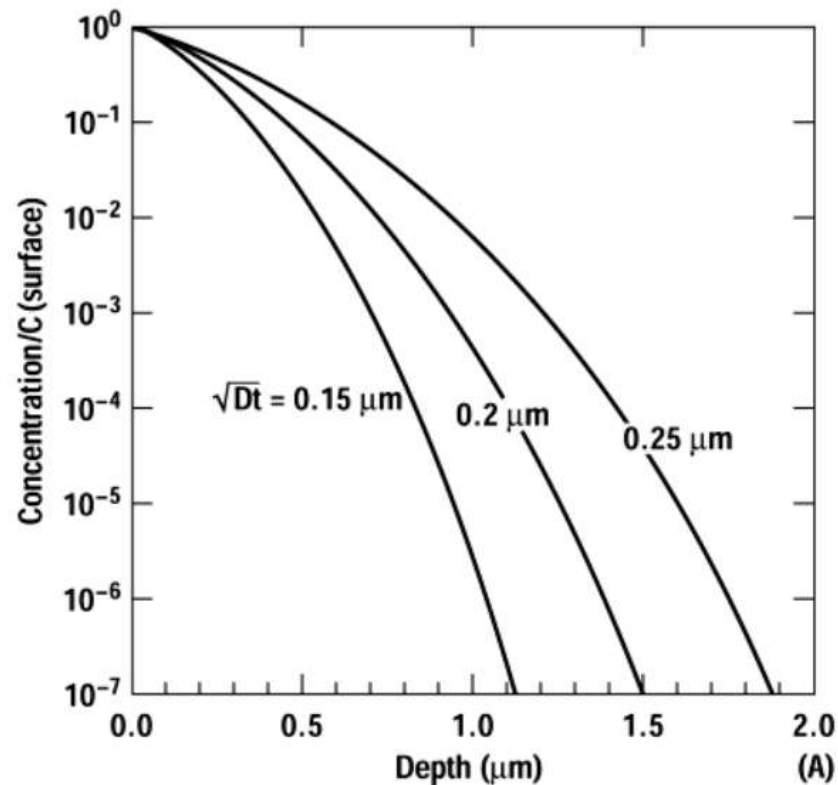
4.3.4 Two step Dopant Diffusion



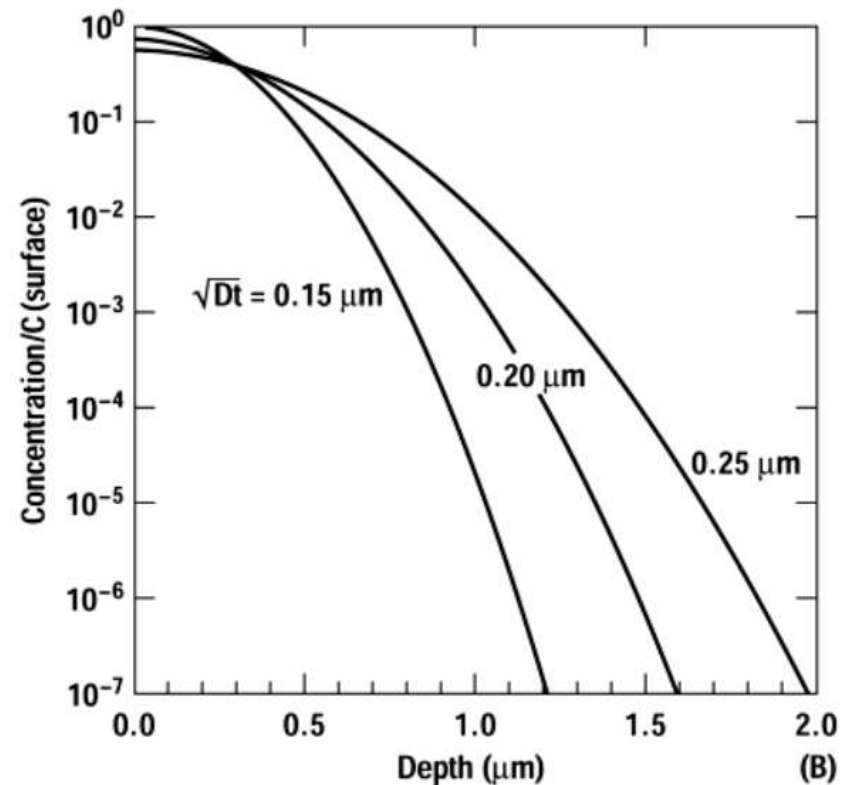
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Normalized Concentration versus Depth

Predeposition



Drive-in

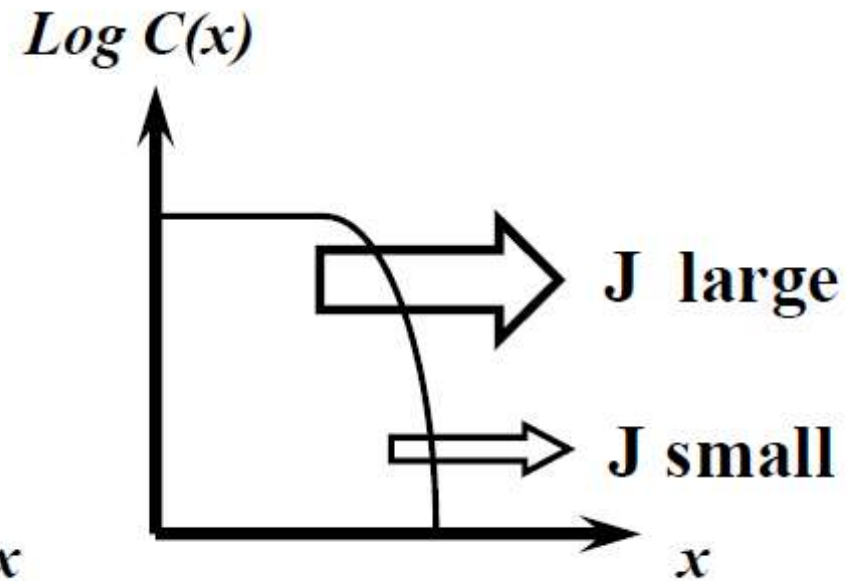
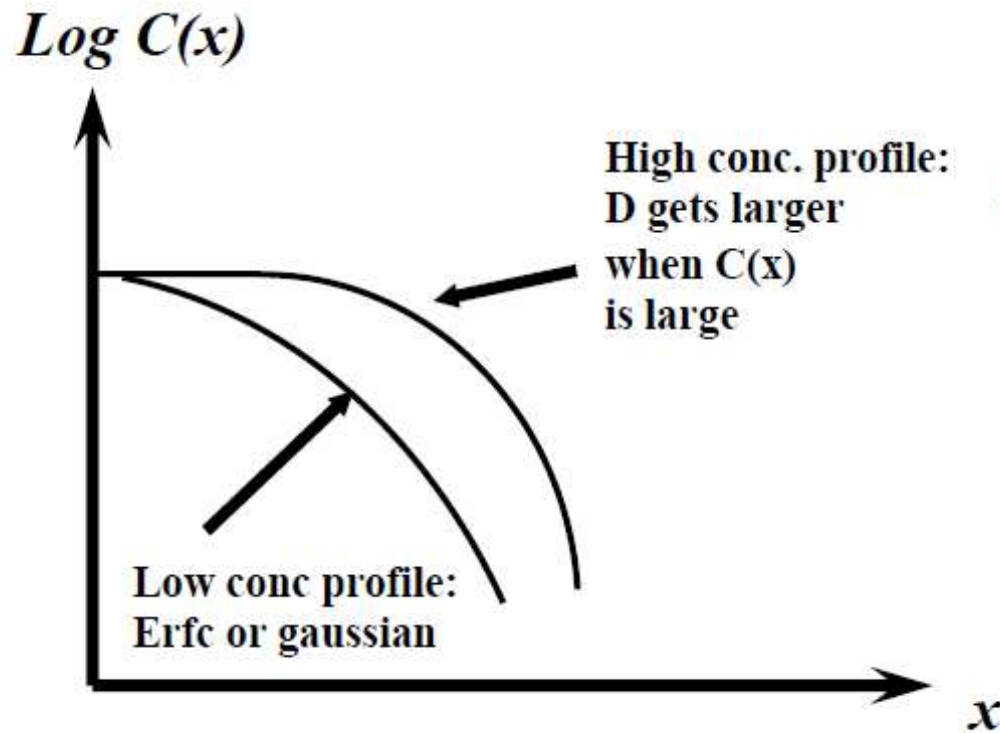


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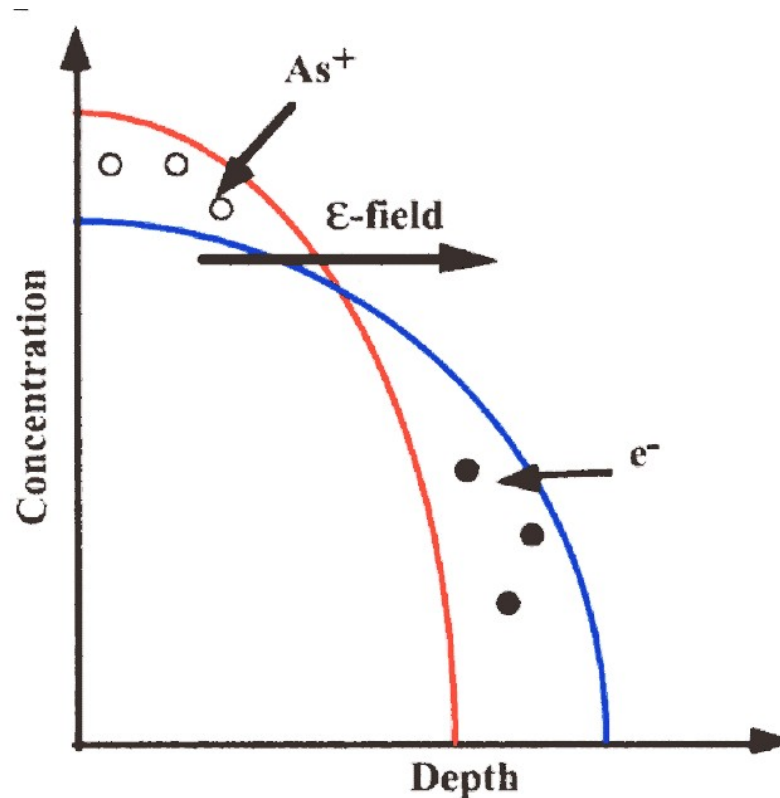
4.4.1 High concentration Diffusion effects



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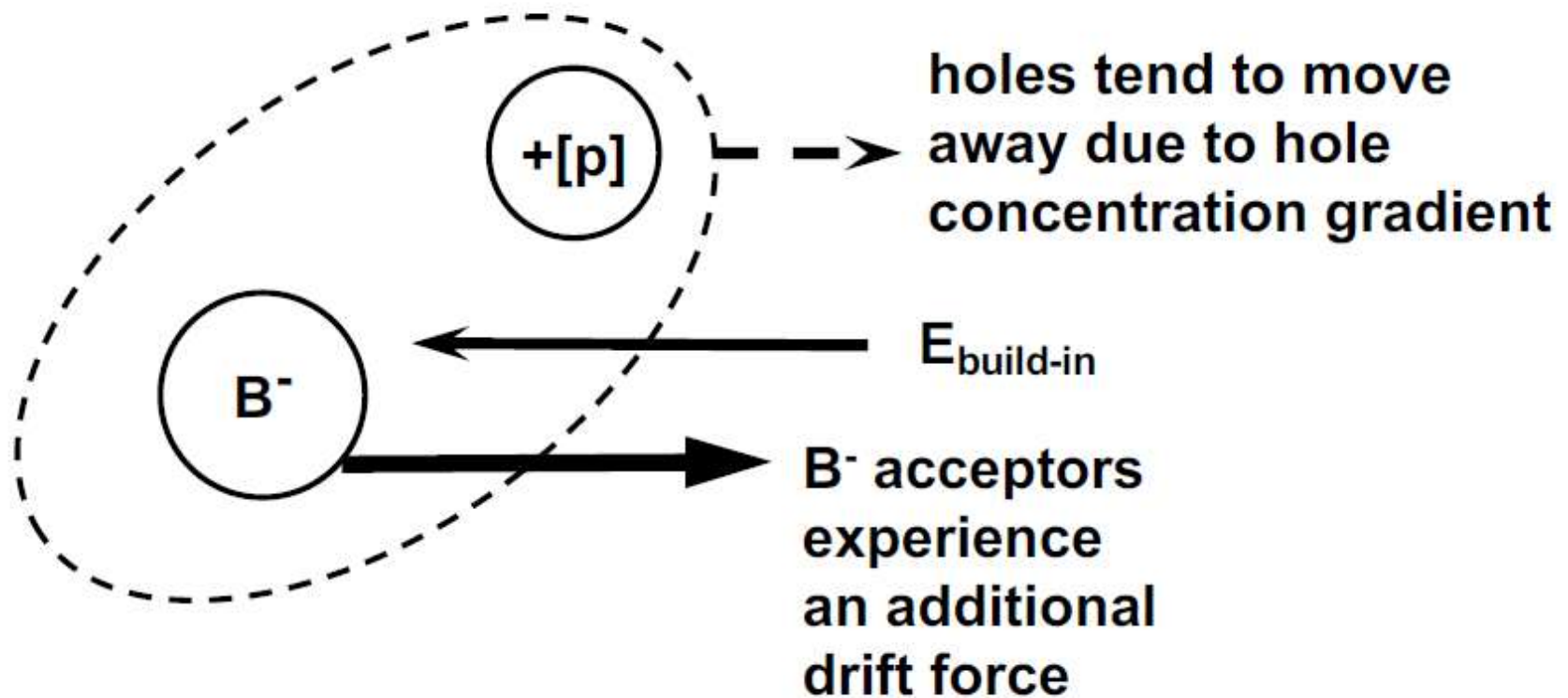


When the doping is higher than n_i , Electric field effects become important



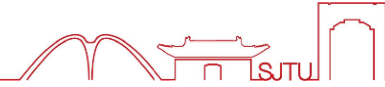
Electrons or holes diffuse ahead of the dopants until they reach a steady-state condition where the drift flux from the internal electric field balances the electron diffusion flux.

4.4.2 Field enhanced diffusion



Enhanced
Diffusion for B^- acceptor atoms

4.4.3 Modification of Fick's law

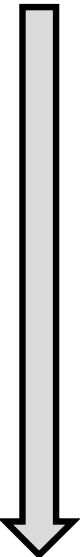


Total impurity atom flux in the presence of both a concentration gradient and an electric field E is

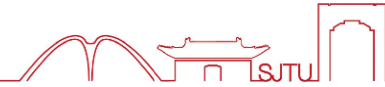
$$F_{total} = F + F' = -\frac{DdC}{dx} + z\mu CE$$

Einstein relation $\mu = D \frac{q}{kT}$

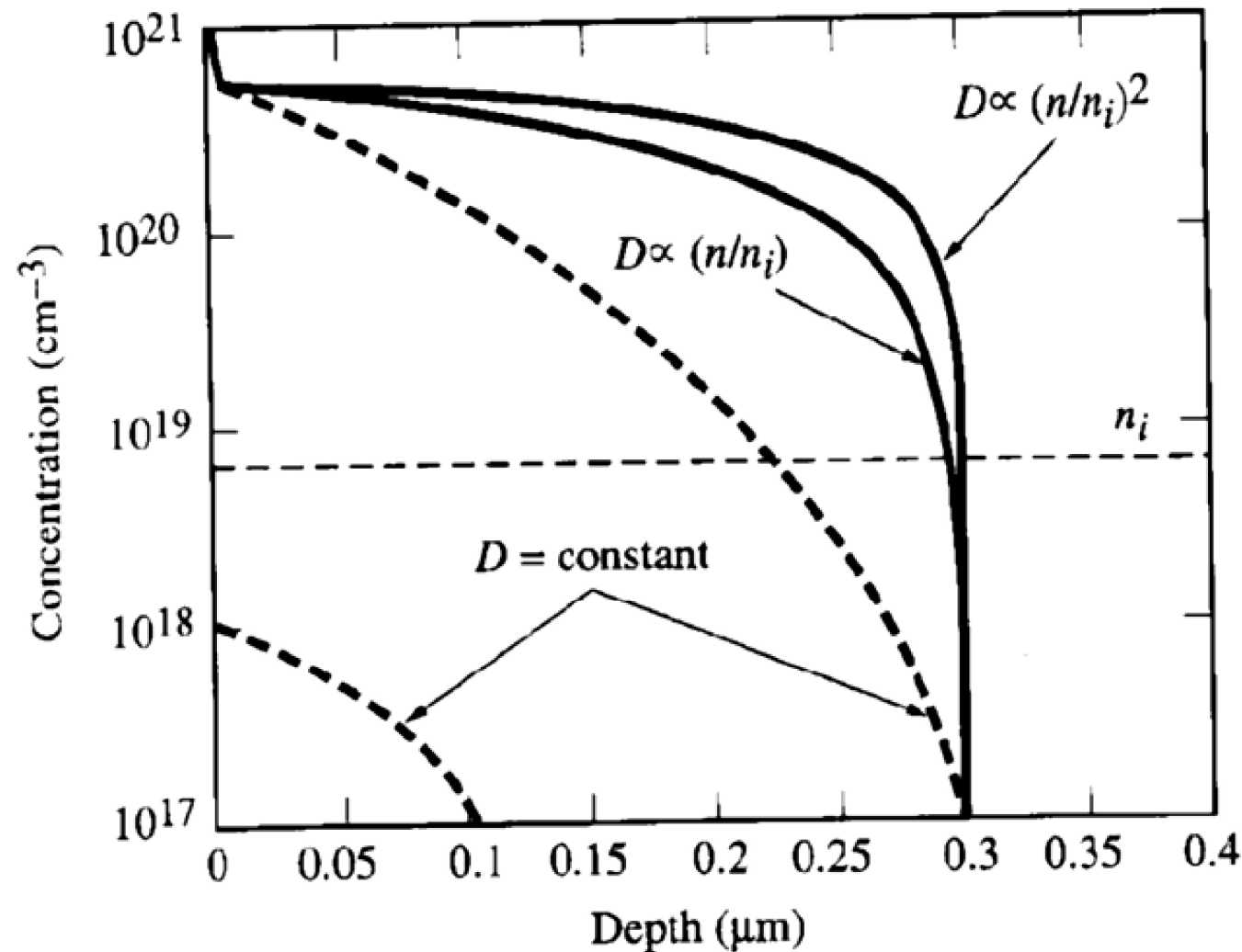
Electric field $E = \frac{\partial \varphi}{\partial x}$ $\varphi = -\frac{kT}{q} \ln \frac{n}{n_i}$


$$F_{total} = -\frac{DdC}{dx} - DC \frac{\partial}{\partial x} \ln \frac{n}{n_i}$$

4.4.4 Dopants profiles



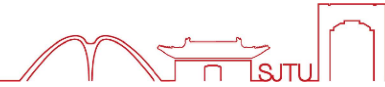
- Example : High Concentration Arsenic diffusion profile becomes “box--like”



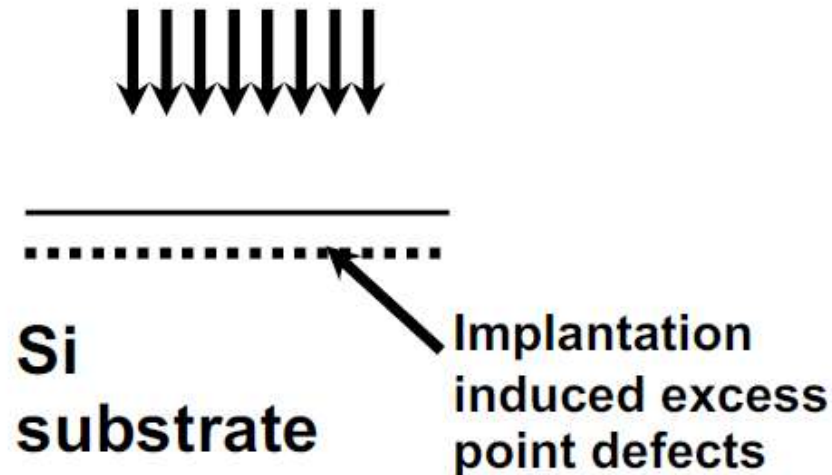
- If doping conc $< n_i$:
 - Use constant diffusivity solutions
 - (profile is erfc or half-gaussian)
- If doping conc $> n_i$:
 - Solution requires numerical techniques

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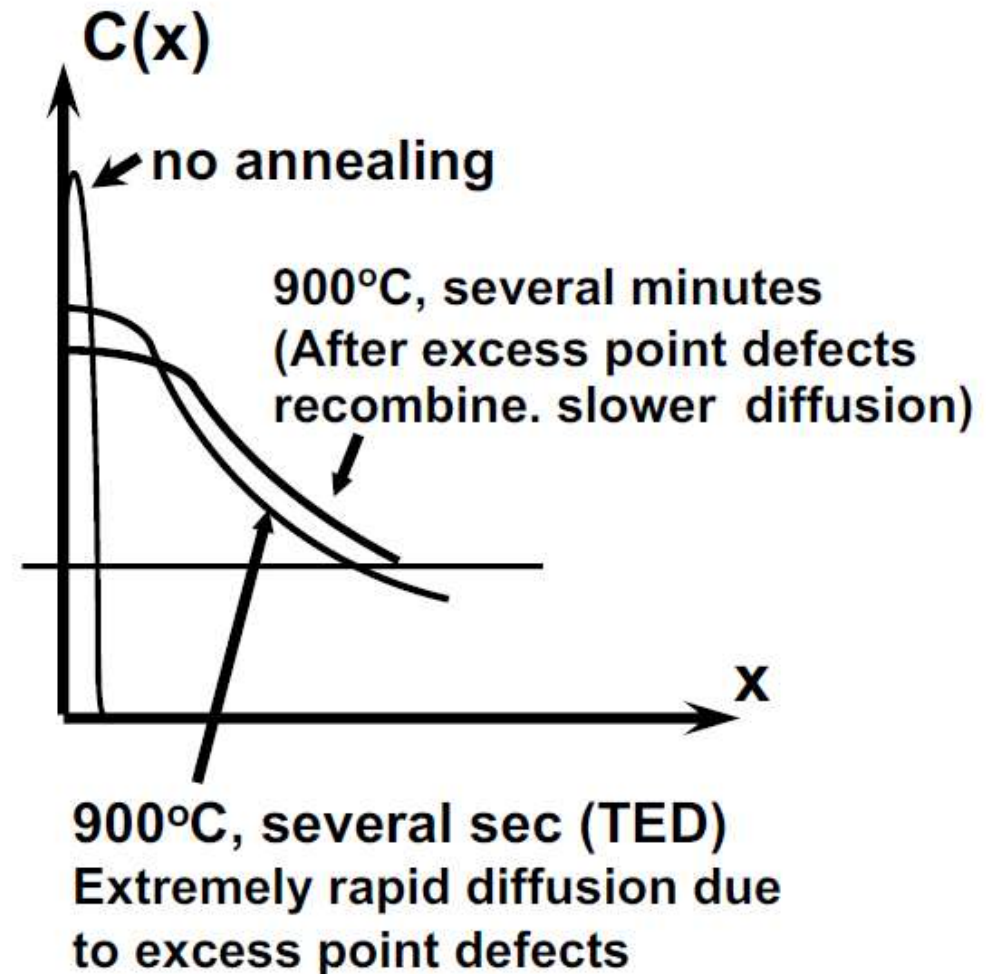
4.5.1 Transient Enhanced Diffusion



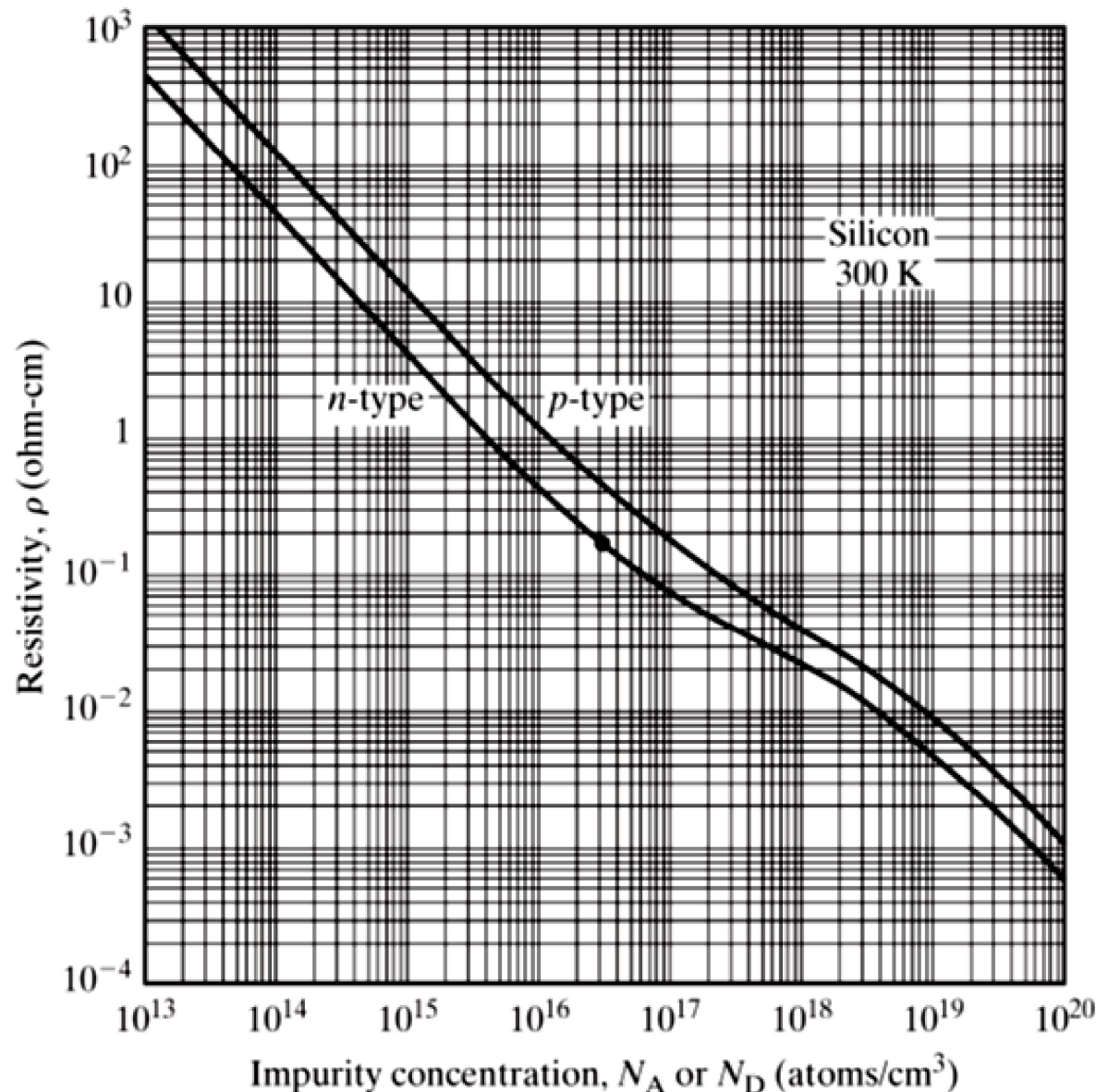
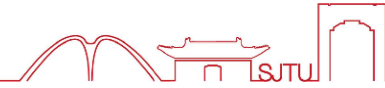
Dopant Implantation



Implantation creates large number of excess Si interstitials and vacancies. (1000X than thermal process). After several seconds of annealing, the excess point defects recombine.



4.5.2 Resistivity Vs. Doping



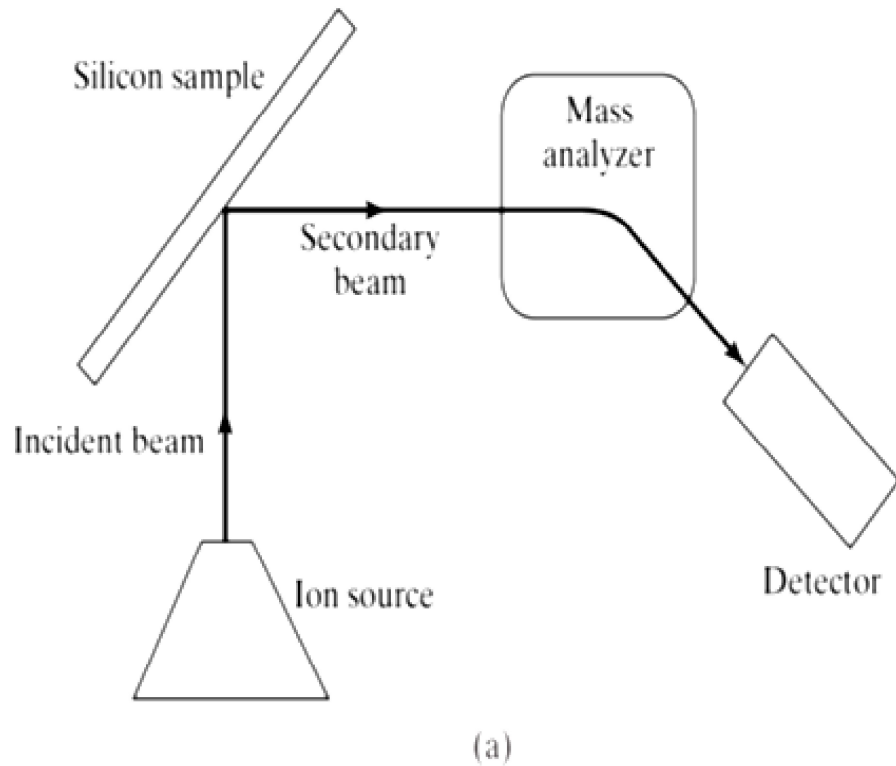
$$\rho = \sigma^{-1} = [q(\mu_n n + \mu_p p)]^{-1}$$

$$\text{n-type} : \rho \cong [q\mu_n(N_D - N_A)]^{-1}$$

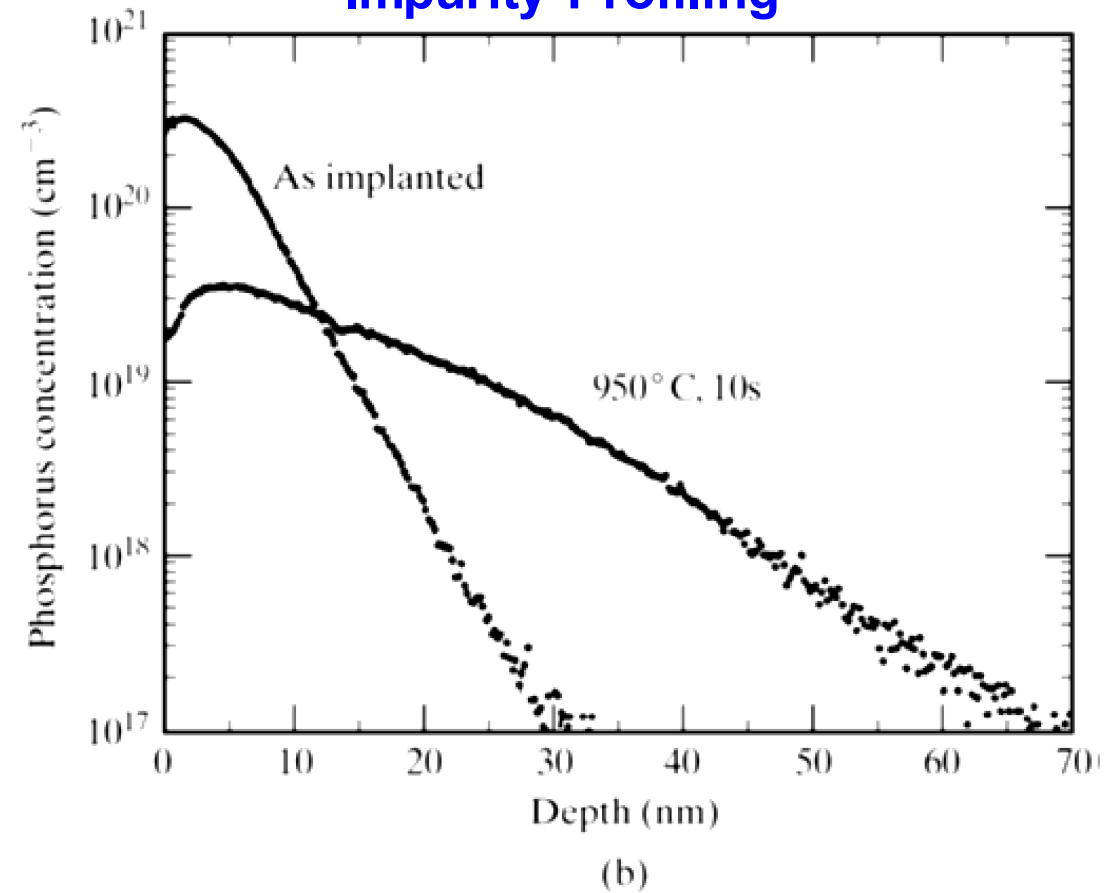
$$\text{p-type} : \rho \cong [q\mu_p(N_A - N_D)]^{-1}$$

4.5.3 SIMS for impurity profiles

Secondary Ion Mass Spectroscopy (SIMS)



Impurity Profiling



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➤ Why is diffusion important?

Determine impurity concentration and resistivity

➤ Basics of diffusion

Definition, solubility, mechanism

➤ Fick's law for diffusion

$$F = - \frac{DdC}{dx} \quad \frac{dC}{dt} = \frac{Dd^2C}{dx^2}$$

First and second law

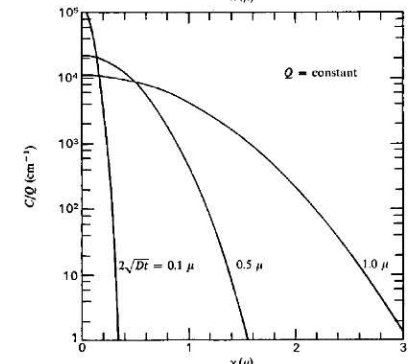
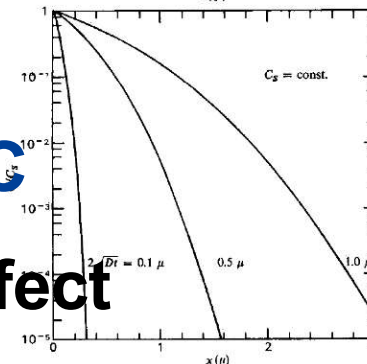
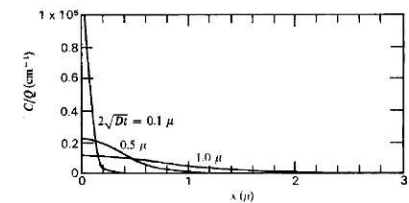
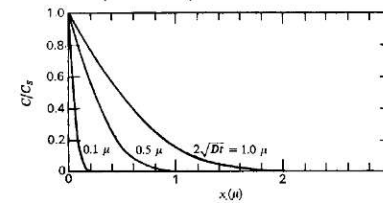
➤ Two types of diffusion

Gaussian and error function

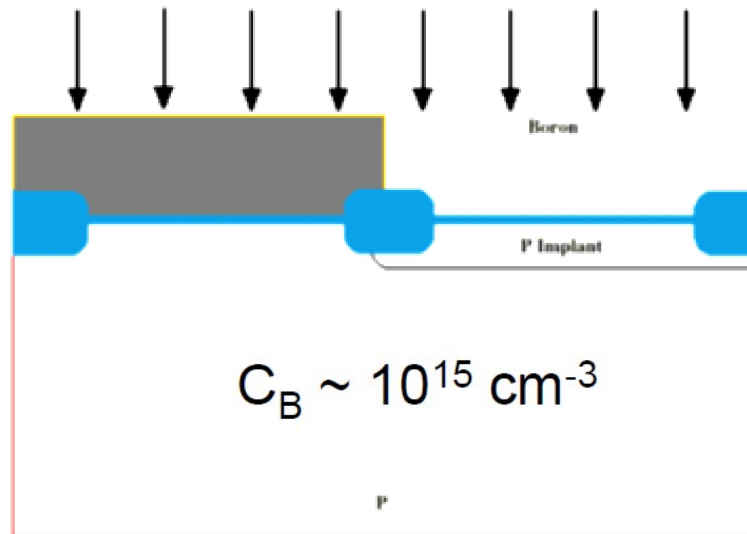
Predeposition, drive-in in actual IC

➤ High concentration diffusion effect

Why? How?



Design the Drive-in Conditions (i.e. the temperature and time) for a CMOS P-well



**Want sheet resistance
=900 ohm/square
And $X_j = 3 \text{ um}$**

The sheet resistance of a shallow junction with uniform doping

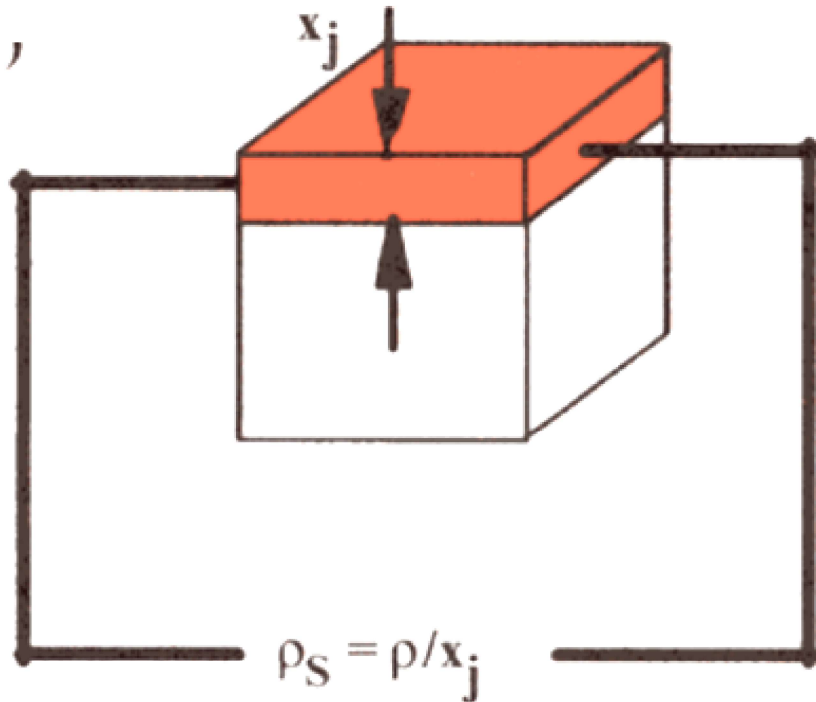
$$\mathbf{R} = \frac{\rho}{x_j} \Omega / \mathbf{Square} \equiv \rho_s$$

In general: $\rho(x) = \frac{1}{q n(x) \mu[n(x)]}$
(ohm-cm)

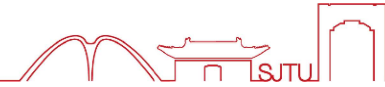
For a non-uniformly doped layer

$$\rho_s = \frac{\bar{\rho}}{x_j} = \frac{1}{q \int_0^{x_j} [n(x) - C_B] \mu[n(x)] dx}$$

This equation has been integrated numerically (by Irvin) for several analytical profiles (e.g. Gaussian and erfc profiles)



4.6.2 Homework



$$\rho_S = 900 \text{ } \Omega/\text{square}$$

$$x_j = 3 \text{ } \mu\text{m}$$

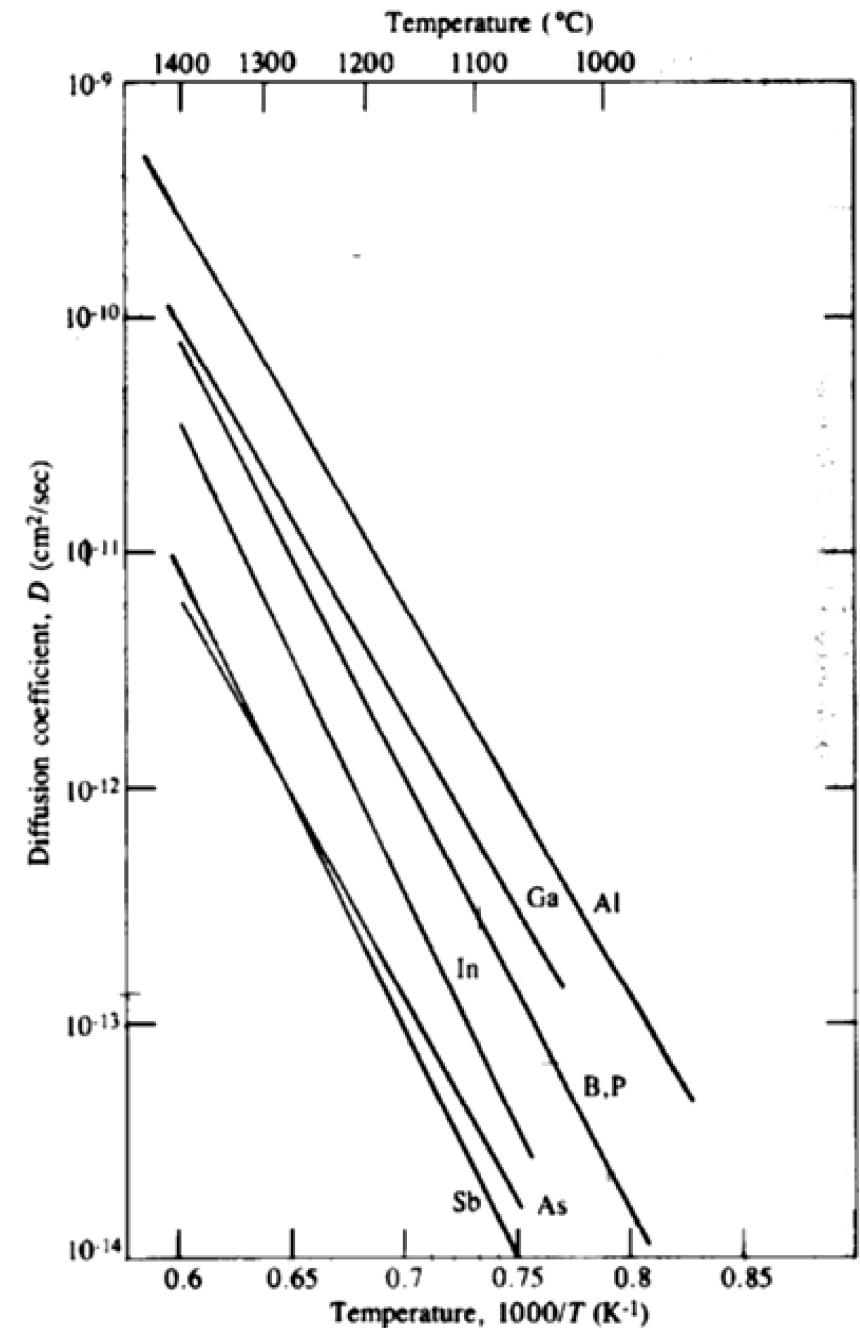
$$C_B = 1 \times 10^{15} \text{ cm}^{-3} \text{ (substrate concentration)}$$

- The average conductivity of the layer is

$$\bar{\sigma} = \frac{1}{\rho_S x_j} = \frac{1}{(900 \Omega/\text{sq})(3 \times 10^{-4} \text{ cm})} = 3.7 (\Omega \cdot \text{cm})^{-1}$$

- From Irvin's curve we obtain (assuming Gaussian)

$$C_S \approx 4 \times 10^{17} / \text{cm}^3$$



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谢 谢

